2024, Mar. 2024 Release LOC:Acute Adult
General Trauma

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Instruction: The differentiation of medical vs. surgical management of trauma patients is complex. From a workflow perspective, InterQual® addresses the surgical management of a trauma patient as follows:

- Patients awaiting surgical intervention for an ambulatory procedure with severe pain requiring hospitalization prior to the procedure may be reviewed at the Observation level of care in the General Surgical subset
- Patients awaiting surgical intervention for a designated inpatient procedure who require pain and/or fluid management may be reviewed at the Acute level of care in the General Surgical subset
- Patients who require initial nonoperative management of a traumatic injury may be reviewed in the General Trauma subset based on the severity of their injury and the interventions being provided
- Patients who require operative management of a traumatic injury following initial medical management should be reviewed in the General Surgical subset
- Criteria have been incorporated in the General Trauma subset to address injuries that require staged operative procedures in addition to medical management (e.g., burns) or require an ambulatory procedure as part of their treatment plan (e.g., eye injuries). In these scenarios, the severity of their injury is the driver for admission and not the surgical intervention.

Patients who present with multiple traumatic injuries should be reviewed based on the injury that is of highest acuity or clinical importance (e.g., requires the most immediate attention, the highest level of care). As the patient's condition progresses, other injuries or complications may drive continued stay. During each review within the Trauma subset, the patient may be evaluated under a different category (e.g., general, respiratory, neurological), injury, or intervention depending on the continued need for hospitalization. Despite a change in the reason for continued stay, the review should be conducted for the next Episode Day, rather than starting from Episode Day 1, since the injury originated from the initial trauma.
In addition to the general complications of trauma (e.g., pain, cardiac arrest), criteria for the following conditions are found within this subset:

- Abuse or neglect
- Accidental hypothermia
- Aortic injury
- Burns
- Decompression sickness
- Drowning
- Fall (review based on injury)

- Head injury
- High-risk trauma (e.g., fall > 10 ft, motor vehicle, pedestrian or bicyclist, rider separated from transport vehicle)
- Inhalation injury
- Electrical injury
- Eye injury requiring surgery
- Foreign body
- Fracture (including chest wall fractures)
- Organ injury
- Snakebite
- Thoracic injury
- Vascular injury

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)
Select Day, One:
- **Initial review, One:** [1, 2]
  (From arrival in ED through decision to admit)
  (Symptom or finding within 24h)
  - **OBSERVATION,** ≥ **One:** [3]
    - **General,** ≥ **One:**
      - Abuse or neglect, actual or suspected resulting in injury [4, 5]
      - Foreign body and, **One:**
        - Unable to extract in ED [6]
        - Post extraction monitoring at least every 4h [7, 8]
      - Pain, severe and unresponsive to ≥ 2 doses of analgesic prior to admit decision
      - Snakebite and, **All:** [9]
        - Assess site for ecchymosis and swelling at least every 4h [10]
        - Coagulation studies [11]
        - Neurological assessment at least 4x/24h [12]
        - Neurovascular assessment at least 4x/24h [13]
    - **Cardiovascular or peripheral vascular, One:**
      - Chest trauma and, **Both:** [14, 15, 16]
        - Initial ECG normal or unchanged
        - Initial troponin normal or at baseline [17]
    - **Gastrointestinal, Genitourinary, or biliary,** ≥ **One:**
      - Abdominal trauma and pregnancy, not in active labor and, **Both:** [18, 19]
        - Fetal heart rate monitoring [20]
        - Uterine monitoring [21]
      - Organ injury, suspected [22, 23]
    - **Musculoskeletal,** ≥ **One:**
      - Chest wall fracture, ≥ **One:**
        - First rib fracture [24]
        - Sternal fracture [25]
        - Underlying COPD or heart failure and chest wall fracture [26, 27]
      - Fracture, subluxation, or dislocation, **Both:**
        - Finding, ≥ **One:**
          - Facial fracture
          - Fracture, subluxation, or dislocation of spine [28]
          - Fracture of ankle, femur, hip or pelvis
          - Dislocation of major joint prosthesis
        - Surgical consultation ≤ 24h [29]
    - **Neurological, One:**
      - Head trauma and negative head CT and, ≥ **One:** [30, 31]
        - GCS 13-14 and < baseline [17, 32, 33]
        - Anticoagulant or antiplatelet use [34]
        - Concussion within 4 days and persistent symptoms [35, 36]
        - Difficult neurological assessment, ≥ **One:** [37, 38]
          - Cognitive impairment
          - Drug or alcohol intoxication
          - Neurodegenerative movement disorder
    - **Respiratory,** ≥ **One:**
      - Pneumothorax and, ≥ **One:** [39, 40, 41]
        - Intervention, **Both:**
          - Small bore chest tube placed [42, 43]
          - Oxygen requirement back to baseline post chest tube placement [17]
        - Oximetry monitoring and repeat CXR performed within 24h

- Thoracic injury, **Both:**
  - Finding, ≥ **One:**
    - Hemothorax [44]
    - Pneumomediastinum
  - Oximetry monitoring and repeat CXR performed within 24h
- **Skin, One:**
  - 2nd degree burn and, **Both:** [45, 46]
    - Patient or caregiver unable to adhere to treatment regimen
    - Outpatient services unavailable [47]

- **ACUTE, ≥ One:** [48]
  - **General, ≥ One:**
    - Bleeding and anticoagulation reversal [49]
    - Traumatic injury and frailty, **Both:** [50, 51, 52]
      - Age ≥ 65 yrs
      - Finding, ≥ **One:**
        - Frailty, active diagnosis [53]
        - Frailty characteristics, ≥ **Two:** [54]
          - Chronic condition(s), ≥ **One:**
            - Dementia or cognitive impairment [55]
            - Multimorbidity, ≥ 2 chronic illnesses [56, 57]
          - Functional impairment [58, 59]
          - Malnutrition [60]
          - Polypharmacy, ≥ 5 chronic medications [61, 62]
  - **Cardiovascular or peripheral vascular, One:**
    - Vascular compromise, **Both:**
      - Finding, ≥ **One:**
        - Compartment syndrome, suspected [63]
        - Limb ischemia, suspected
      - Neurovascular assessment at least 6x/24h, ≤ 2d [13]
  - **Eye, One:**
    - Eye injury and surgery (includes ambulatory procedures) performed within 24h [64, 65, 66]
  - **Gastrointestinal, Genitourinary, or biliary, One:**
    - Abdominal trauma and organ injury, actual [67, 68, 69]
  - **Musculoskeletal, ≥ One:**
    - Chest wall fracture and, ≥ **One:** [26]
      - Analgesic, continuous
      - Analgesic, regional ≤ 3d [70]
      - Respiratory impairment or high risk, ≥ **One:** [71]
        - Age > 65 yrs and > 2 rib fractures [72]
        - Dyspnea [73]
        - Incentive spirometry < 1000 mL
        - O_2 sat ≤ 91%(0.91) and < baseline [17]
    - Facial fracture (includes reduction and immobilization performed in the OR) and, ≥ **One:**
      - CSF leak and neurological assessment at least 4x/24h, ≤ 7d [12, 74]
      - Inability to take PO and IV fluid [75]
      - Risk for airway compromise and oximetry monitoring every 4h, ≤ 2d [76]
    - Fracture, subluxation, or dislocation, **Both:**
      - Finding, ≥ **One:**
        - Dislocation of major joint prosthesis
        - Fracture, subluxation, or dislocation of spine [28, 77]
        - Fracture of femur, hip, or pelvis
      - Intervention, ≥ **One:**
        - Analgesic ≥ 4x/24h or continuous

- Analgesic, regional ≤ 3d [78]
- Blood product transfusion [79]
- Pending medical clearance for surgery ≤ 24h
- Traction ≤ 4d (includes OR placement) [80]

- ⊖ **Neurological, One:**
  - ⊖ Head trauma and, ≥ **One:**
    - Basilar skull fracture [81]
    - CSF leak
    - Negative head CT and GCS 9-12 and < baseline [17, 31, 32, 82]
- ⊖ **Respiratory, ≥ One:**
  - ⊖ Chest tube placement and, ≥ **One:** [39, 41, 83]
    - Large bore chest tube placed [42, 43]
    - Suction, continuous
    - O_2 requirement > baseline post chest tube placement [17]
    - Video-assisted thoracoscopic surgery
  - ⊖ Drowning and, ≥ **One:** [84]
    - Rales on exam and oximetry monitoring every 4h
    - Requiring supplemental oxygen [85]
  - Pulmonary contusion and requiring supplemental oxygen [85, 86]
- ⊖ **Skin, Both:**
  - ⊖ Burn, ≥ **One:** [45, 87]
    - ⊖ 2nd degree burn (partial-thickness) and, ≥ **One:** [88]
      - ≥ 5%(0.05) BSA and age > 50 yrs [89]
      - ≥ 10%(0.10) BSA
      - ⊖ Affected site, ≥ **One:**
        - Face
        - Feet
        - Genitalia
        - Hands
        - Joint
    - 3rd degree burn (full-thickness)
    - Chemical [90]
    - Circumferential (excludes 1st degree burn)
    - Electrical burn
    - Inhalation injury [91]
  - ⊖ Burn therapy, ≥ **One:**
    - Complex wound care [92, 93]
    - ⊖ Hydration and nutrition, ≥ **One:** [94]
      - Enteral feeds
      - IV fluid [75]
      - TPN
    - ⊖ Medication, ≥ **One:**
      - Analgesic ≥ 4x/24h or continuous [95]
      - Sedative ≥ 3x/24h [96]
    - Neurovascular assessment at least 6x/24h [13, 97]
    - O_2 sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen [17, 85]

⊖ **INTERMEDIATE, ≥ One:** [98]

- ⊖ Accidental hypothermia, **One:** [99, 100]
  - Temperature 89.6°F to 95.0°F(32.0°C to 35.0°C) PR
- ⊖ Chest trauma and, ≥ **One:** [14, 101]
  - ECG with new abnormality
  - Troponin > ULN and > baseline [17]
- Decompression sickness [102, 103]

- ⊖ High risk trauma, ≥ **One:** (104)
  - Fall > 10 ft(3.0 m)
  - ⊖ Motor vehicle trauma and, ≥ **One:**
    - Death in passenger compartment
    - Ejection from vehicle, partial or complete
    - ⊖ Intrusion into vehicle (including roof), ≥ **One:**
      - > 12 inches(30.4 cm) occupant site
      - > 18 inches(45.7 cm) other site
    - Need for extrication for entrapped patient
    - Vehicle telemetry data consistent with high risk for injury (105)
  - ⊖ Pedestrian or bicyclist and, ≥ **One:**
    - Thrown
    - Run over
    - Impact > 20 mph(32.2 kph)
  - Rider separated from transport vehicle (106)
- HFNC and respiratory monitoring every 3-4h (107, 108)
- NIPPV (109)
- Oxygen ≥ 40%(0.40) (85)
- Retroperitoneal hematoma or vascular contusion and Hct or Hb monitoring at least 4x/24h (110)

⊖ **CRITICAL,** ≥ **One:** (111)

- Abdominal trauma and angioembolization ≤ 24h (112)
- ⊖ Accidental hypothermia, **One:** (99, 100)
  - Temperature < 89.6°F(32.0°C) PR
  - Temperature ≤ 95.0°F(35.0°C) PR and hemodynamic instability (113)
- Anticonvulsant, continuous
- ⊖ Aortic injury and, ≥ **One:** (114)
  - Antihypertensive, continuous
  - Endovascular repair ≤ 24h
  - Hct or Hb monitoring at least 4x/24h, ≤ 2d
- ⊖ Burn, **Both:** (45, 87)
  - ⊖ Finding, ≥ **One:**
    - 2nd degree (partial-thickness) or 3rd degree burn (full-thickness) (88)
    - Chemical (90)
    - Circumferential
    - Electrical burn (115)
  - ⊖ Intervention, ≥ **One:**
    - Anxiolytic, continuous (116)
    - IV fluid resuscitation and urine output monitoring every 1-2h (117)
    - Neurovascular assessment every 1-2h (13, 97)
    - Wound care requiring procedural sedation (92, 118, 119)
- Cardiac arrest and post resuscitation care ≤ 2d (120)
- ⊖ Cerebral edema and, ≥ **One:**
  - ICP monitoring (includes OR placement) (121)
  - ⊖ Hyperosmolar therapy, ≥ **One:** (122)
    - Mannitol
    - Hypertonic saline ≥ 3%(0.03) solution
- DIC and blood product transfusion (123)
- ECMO or ECLS (124)
- ⊖ Electrical injury and, ≥ **One:** (125)
  - Arrhythmia (126)
  - ECG with new or worsening ischemia (127, 128)
  - High voltage (129)
  - Loss of consciousness

- Troponin > ULN and > baseline [17]
- Emergent tracheostomy or cricothyrotomy ≤ 24h [130]
- External ventricular or lumbar drainage (includes OR placement)
- Flexible bronchoscopy ≥ 2x/24h [131]
- Hemodynamic instability, **All:** [132]
  - Initial treatment, ≥ **One:**
    - ≥ 2 units blood product transfusion
    - ≥ 2 liters IV fluid [133]
    - ≥ 1 unit blood product transfusion and ≥ 1 liter IV fluid
  - Findings after initial treatment, ≥ **One:**
    - Heart rate > 120/min
    - Systolic blood pressure < 100 mmHg and < baseline [17]
    - MAP < 65 mmHg
  - Requiring ongoing resuscitation
- Hemodynamic monitoring, invasive [134]
- HFNC and respiratory monitoring every 1-2h [107, 108]
- IABP [135]
- Mechanical ventilation
- Nebulizer or MDI treatment, ≥ **One:** [136]
  - Every 1-2h
  - Continuous
- Neurological assessment every 1-2h and, ≥ **One:** [12]
  - Brain hematoma, contusion, or hemorrhage [137]
  - Coma, stupor, or obtundation or GCS ≤ 8 [32]
  - Depressed skull fracture [138]
  - Focal neurologic deficits, ≥ **One:**
    - Aphasia
    - Ataxia
    - Blindness
    - Diplopia
    - Dysarthria
    - Dysphasia
    - Paresis or paralysis of extremities [139]
    - Visual field loss [140]
  - Spinal cord injury [141]
- NIPPV and, ≥ **One:** [109]
  - Arterial Po_2 ≤ 39 mmHg(5.2 kPa)
  - Arterial or venous Pco_2 ≥ 60 mmHg(8.0 kPa) and, **One:**
    - COPD and arterial or venous pH ≤ 7.24
    - No hx of COPD
- Organ injury, ≥ **One:** [142]
  - Grade ≥ III liver, pancreas, or spleen injury [143]
  - Grade ≥ IV kidney injury [144]
- Pelvic fracture and, ≥ **One:** [145]
  - Angioembolization ≤ 24h
  - Preperitoneal pelvic packing [146]
  - Pelvic external fixation [147]
- Snakebite and, ≥ **One:** [148]
  - Antivenom [149]
  - Neurovascular assessment every 1-2h [13]
  - Neurological assessment every 1-2h [12]
- Urgent electrical cardioversion performed within last 24h
- Vascular compromise, **Both:**

- Finding, ≥ **One:**
  - Compartment syndrome, actual or suspected [63]
  - Limb ischemia, actual or suspected
- Intervention, ≥ **One:**
  - Endovascular revascularization ≤ 24h [150]
  - Neurovascular assessment every 1-2h [13]
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

Episode Day 1, One: [1]

(Symptom or finding within 24h)

- **OBSERVATION, ≥ One:** [3]
  - **General, ≥ One:**
    - Abuse or neglect, actual or suspected resulting in injury and, **Both:** [4, 5]
      - Social work or case management consult [151]
      - Notification of adult protective services [152]
    - Foreign body and, **One:**
      - Unable to extract in ED [6]
      - Post extraction monitoring at least every 4h [7, 8]
    - Pain, severe, **Both:**
      - Unresponsive to ≥ 2 doses of analgesic prior to admit decision
      - Requiring additional analgesic ≥ 2 doses after admission
    - Snakebite and, **All:** [9]
      - Assess site for ecchymosis and swelling at least every 4h [10]
      - Coagulation studies [11]
      - Neurological assessment at least 4x/24h [12]
      - Neurovascular assessment at least 4x/24h [13]
  - **Cardiovascular or peripheral vascular, One:**
    - Chest trauma and, **All:** [14, 15, 16]
      - Initial ECG normal or unchanged
      - Initial troponin normal or at baseline [17]
      - Continuous cardiac monitoring (excludes Holter)
  - **Gastrointestinal, Genitourinary, or biliary, ≥ One:**
    - Abdominal trauma and pregnancy, not in active labor and, **Both:** [18, 19]
      - Fetal heart rate monitoring [20]
      - Uterine monitoring [21]
    - Organ injury, suspected and, ≥ **One:** [22, 23]
      - Hct or Hb monitoring at least 4x/24h
      - Repeat imaging scheduled within 24h
      - Abdominal exam at least 2x/24h
  - **Musculoskeletal, ≥ One:**
    - Chest wall fracture, **Both:**
      - Finding, ≥ **One:**
        - First rib fracture [24]
        - Sternal fracture [25]
        - Underlying COPD or heart failure and chest wall fracture [26, 27]
      - Oximetry monitoring every 4h
    - Fracture, subluxation, or dislocation, **Both:**
      - Finding, ≥ **One:**
        - Facial fracture
        - Fracture, subluxation, or dislocation of spine [28]
        - Fracture of ankle, femur, hip, or pelvis
        - Dislocation of major joint prosthesis
      - Surgical consultation ≤ 24h [29]
  - **Neurological, One:**

- Head trauma and negative head CT and, **Both:** [30, 31]
  - Finding, ≥ **One:**
    - GCS 13-14 and < baseline [17, 32, 33]
    - Anticoagulant or antiplatelet use [34]
    - Concussion within 4 days and persistent symptoms [35, 36]
    - Difficult neurological assessment, ≥ **One:** [37, 38]
      - Cognitive impairment
      - Drug or alcohol intoxication
      - Neurodegenerative movement disorder
  - Neurological assessment at least 4x/24h [12]
- **Respiratory, ≥ One:**
  - Pneumothorax and, ≥ **One:** [39, 40, 41]
    - Intervention, **Both:**
      - Small bore chest tube placed [42, 43]
      - Oxygen requirement back to baseline post chest tube placement [17]
    - Oximetry monitoring and repeat CXR performed within 24h
  - Thoracic injury, **Both:**
    - Finding, ≥ **One:**
      - Hemothorax [44]
      - Pneumomediastinum
    - Oximetry monitoring and repeat CXR performed within 24h
- **Skin, One:**
  - 2^{nd} degree burn and, **Both:** [45, 46]
    - Patient or caregiver unable to adhere to treatment regimen
    - Outpatient services unavailable [47]

- **ACUTE, ≥ One:** [48]
  - **General, ≥ One:**
    - Bleeding and anticoagulation reversal [49]
    - Traumatic injury and frailty, **All:** [50, 51, 52]
      - Age ≥ 65 yrs
      - Finding, ≥ **One:**
        - Frailty, active diagnosis [53]
        - Frailty characteristics, ≥ **Two:** [54]
          - Chronic condition(s), ≥ **One:**
            - Dementia or cognitive impairment [55]
            - Multimorbidity, ≥ 2 chronic illnesses [56, 57]
          - Functional impairment [58, 59]
          - Malnutrition [60]
          - Polypharmacy, ≥ 5 chronic medications [61, 62]
      - Intervention, ≥ **One:**
        - Analgesic (includes PO) ≥ 2x/24h
        - Continuous cardiac monitoring (excludes Holter)
        - IV fluid [75]
        - Neurological assessment at least 4x/24h [12]
        - Oximetry monitoring every 4h
  - **Cardiovascular or peripheral vascular, One:**
    - Vascular compromise, **Both:**
      - Finding, ≥ **One:**
        - Compartment syndrome, suspected [63]
        - Limb ischemia, suspected
      - Neurovascular assessment at least 6x/24h, ≤ 2d [13]
  - **Eye, One:**
    - Eye injury and surgery (includes ambulatory procedures) performed within 24h [64, 65, 66]

- **Gastrointestinal, Genitourinary, or biliary, One:**
  - Abdominal trauma and, **Both:**
    - Organ injury, actual [67, 68, 69]
    - Intervention, ≥ **One:** [153]
      - Blood product transfusion
      - Hct or Hb monitoring at least 4x/24h
      - NPO and IV fluid [75, 154]
      - Suprapubic catheter, nephrostomy tube, or ureteral stent placed within 24h [155]
- **Musculoskeletal, ≥ One:**
  - Rhabdomyolysis or crush syndrome (**use Rhabdomyolysis or Crush Syndrome subset**)
  - Chest wall fracture and, ≥ **One:** [26]
    - Analgesic, continuous
    - Analgesic, regional ≤ 3d [70]
    - Respiratory impairment or high risk, **Both:** [71]
      - Finding, ≥ **One:**
        - Age > 65 yrs and > 2 rib fractures [72]
        - Dyspnea [73]
        - Incentive spirometry < 1000 mL
        - O_2 sat ≤ 91%(0.91) and < baseline [17]
      - Intervention, ≥ **One:**
        - Supplemental oxygen [85]
        - Respiratory interventions at least 6x/24h, ≤ 4d [156]
  - Facial fracture (includes reduction and immobilization performed in the OR) and, ≥ **One:**
    - CSF leak and neurological assessment at least 4x/24h, ≤ 7d [12, 74]
    - Inability to take PO and IV fluid [75]
    - Risk for airway compromise and oximetry monitoring every 4h, ≤ 2d [76]
  - Fracture, subluxation, or dislocation, **Both:**
    - Finding, ≥ **One:**
      - Dislocation of major joint prosthesis
      - Fracture, subluxation, or dislocation of spine [28, 77]
      - Fracture of femur, hip, or pelvis
    - Intervention, ≥ **One:**
      - Analgesic ≥ 4x/24h or continuous
      - Analgesic, regional ≤ 3d [78]
      - Blood product transfusion [79]
      - Pending medical clearance for surgery ≤ 24h
      - Traction ≤ 4d (includes OR placement) [80]
- **Neurological, One:**
  - Head trauma and, **Both:**
    - Finding, ≥ **One:**
      - Basilar skull fracture [81]
      - CSF leak [157]
      - Negative head CT and GCS 9-12 and < baseline [17, 31, 32, 82]
    - Neurological assessment at least 4x/24h [12]
- **Respiratory, ≥ One:**
  - Chest tube placement and, ≥ **One:** [39, 41, 83]
    - Large bore chest tube placed [42, 43]
    - Suction, continuous
    - O_2 requirement > baseline post chest tube placement [17]
    - Video-assisted thoracoscopic surgery
  - Drowning and, ≥ **One:** [84]
    - Rales on exam and oximetry monitoring every 4h
    - Requiring supplemental oxygen [85]

- Pulmonary contusion and requiring supplemental oxygen [85, 86]
- **Skin, Both:**
  - Burn, ≥ **One:** [45, 87]
    - 2^{nd} degree burn (partial-thickness) and, ≥ **One:** [88]
      - ≥ 5%(0.05) BSA and age > 50 yrs [89]
      - ≥ 10%(0.10) BSA
      - Affected site, ≥ **One:**
        - Face
        - Feet
        - Genitalia
        - Hands
        - Joint
    - 3^{rd} degree burn (full-thickness)
    - Chemical [90]
    - Circumferential (excludes 1^{st} degree burn)
    - Electrical burn
    - Inhalation injury [91]
  - Burn therapy, ≥ **One:**
    - Complex wound care [92, 93]
    - Hydration and nutrition, ≥ **One:** [94]
      - Enteral feeds
      - IV fluid [75]
      - TPN
    - Medication, ≥ **One:**
      - Analgesic ≥ 4x/24h or continuous [95]
      - Sedative ≥ 3x/24h [96]
    - Neurovascular assessment at least 6x/24h [13, 97]
    - O_2 sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen [17, 85]
- **INTERMEDIATE, ≥ One:** [98]
  - Accidental hypothermia, **Both:** [99, 100]
    - Temperature 89.6°F to 95.0°F(32.0°C to 35.0°C) PR
    - Active rewarming [158]
  - Chest trauma and, **Both:** [14, 101]
    - Finding, ≥ **One:** [159]
      - ECG with new abnormality
      - Troponin > ULN and > baseline [17]
    - Intervention, **Both:**
      - Continuous cardiac monitoring (excludes Holter)
      - Echocardiogram performed within 24h [160]
  - Decompression sickness and, **One:** [102, 103]
    - Oxygen 100%(1.00) [85]
    - Hyperbaric oxygen therapy (HBOT)
  - High risk trauma, **Both:** [104]
    - Finding, ≥ **One:**
      - Fall > 10 ft(3.0 m)
      - Motor vehicle trauma and, ≥ **One:**
        - Death in passenger compartment
        - Ejection from vehicle, partial or complete
        - Intrusion into vehicle (including roof), ≥ **One:**
          - > 12 inches(30.4 cm) occupant site
          - > 18 inches(45.7 cm) other site
        - Need for extrication for entrapped patient
        - Vehicle telemetry data consistent with high risk for injury [105]

- ⊖ Pedestrian or bicyclist and, ≥ **One:**
  - Thrown
  - Run over
  - Impact > 20 mph(32.2 kph)
- Rider separated from transport vehicle [106]
- Neurological assessment every 3-4h [12]
- HFNC and respiratory monitoring every 3-4h [107, 108]
- NIPPV [109]
- Oxygen ≥ 40%(0.40) [85]
- Retroperitoneal hematoma or vascular contusion and Hct or Hb monitoring at least 4x/24h [110]

⊖ **CRITICAL, ≥ One:** [111]

- Rhabdomyolysis or crush syndrome (**use Rhabdomyolysis or Crush Syndrome subset**)
- Abdominal trauma and angioembolization ≤ 24h [112]
- ⊖ Accidental hypothermia, **Both:** [99, 100]
  - ⊖ Finding, **One:**
    - Temperature < 89.6°F(32.0°C) PR
    - Temperature ≤ 95.0°F(35.0°C) PR and hemodynamic instability [113]
  - Active rewarming [158]
- Anticonvulsant, continuous
- ⊖ Aortic injury and, ≥ **One:** [114]
  - Antihypertensive, continuous
  - Endovascular repair ≤ 24h
  - Hct or Hb monitoring at least 4x/24h, ≤ 2d
- ⊖ Burn, **Both:** [45, 87]
  - ⊖ Finding, ≥ **One:**
    - 2nd degree (partial-thickness) or 3rd degree burn (full-thickness) [88]
    - Chemical [90]
    - Circumferential
    - Electrical burn [115]
  - ⊖ Intervention, ≥ **One:**
    - Anxiolytic, continuous [116]
    - IV fluid resuscitation and urine output monitoring every 1-2h [117]
    - Neurovascular assessment every 1-2h [13, 97]
    - Wound care requiring procedural sedation [92, 118, 119]
- Cardiac arrest and post resuscitation care ≤ 2d [120]
- ⊖ Cerebral edema and, ≥ **One:**
  - ICP monitoring (includes OR placement) [121]
  - ⊖ Hyperosmolar therapy, ≥ **One:** [122]
    - Mannitol
    - Hypertonic saline ≥ 3%(0.03) solution
- DIC and blood product transfusion [123]
- ECMO or ECLS [124]
- ⊖ Electrical injury and, **Both:** [125]
  - ⊖ Finding, ≥ **One:**
    - Arrhythmia [126]
    - ECG with new or worsening ischemia [127, 128]
    - High voltage [129]
    - Loss of consciousness
    - Troponin > ULN and > baseline [17]
  - Continuous cardiac monitoring (excludes Holter)
- Emergent tracheostomy or cricothyrotomy ≤ 24h [130]
- External ventricular or lumbar drainage (includes OR placement)
- Flexible bronchoscopy ≥ 2x/24h [131]

- ⊖ Hemodynamic instability, **All:** (132)
  - ⊖ Initial treatment, ≥ **One:**
    - ≥ 2 units blood product transfusion
    - ≥ 2 liters IV fluid (133)
    - ≥ 1 unit blood product transfusion and ≥ 1 liter IV fluid
  - ⊖ Findings after initial treatment, ≥ **One:**
    - Heart rate > 120/min
    - Systolic blood pressure < 100 mmHg and < baseline (17)
    - MAP < 65 mmHg
  - Requiring ongoing resuscitation
- Hemodynamic monitoring, invasive (134)
- HFNC and respiratory monitoring every 1-2h (107, 108)
- IABP (135)
- Mechanical ventilation
- ⊖ Nebulizer or MDI treatment, ≥ **One:** (136)
  - Every 1-2h
  - Continuous
- ⊖ Neurological assessment every 1-2h and, ≥ **One:** (12)
  - Brain hematoma, contusion, or hemorrhage (137)
  - Coma, stupor, or obtundation or GCS ≤ 8 (32)
  - Depressed skull fracture (138)
  - ⊖ Focal neurologic deficit, ≥ **One:**
    - Aphasia
    - Ataxia
    - Blindness
    - Diplopia
    - Dysarthria
    - Dysphasia
    - Paresis or paralysis of extremities (139)
    - Visual field loss (140)
  - Spinal cord injury (141)
- ⊖ NIPPV and, ≥ **One:** (109)
  - Arterial Po_2 ≤ 39 mmHg(5.2 kPa)
  - ⊖ Arterial or venous Pco_2 ≥ 60 mmHg(8.0 kPa) and, **One:**
    - COPD and arterial or venous pH ≤ 7.24
    - No hx of COPD
- ⊖ Organ injury, **Both:** (142)
  - ⊖ Finding, ≥ **One:**
    - Grade ≥ III liver, pancreas, or spleen injury (143)
    - Grade ≥ IV kidney injury (144)
  - ⊖ Intervention, ≥ **One:** (153)
    - Hct or Hb monitoring at least 4x/24h
    - Blood product transfusion
- ⊖ Pelvic fracture and, ≥ **One:** (145)
  - Angioembolization ≤ 24h
  - Preperitoneal pelvic packing (146)
  - Pelvic external fixator (147)
- ⊖ Snakebite and, ≥ **One:** (148)
  - Antivenom (149)
  - Neurovascular assessment every 1-2h (13)
  - Neurological assessment every 1-2h (12)
- Urgent electrical cardioversion performed within last 24h
- ⊖ Vascular compromise, **Both:**

- Finding, ≥ **One:**
  - Compartment syndrome, actual or suspected [63]
  - Limb ischemia, actual or suspected
- Intervention, ≥ **One:**
  - Endovascular revascularization ≤ 24h [150]
  - Neurovascular assessment every 1-2h [13]
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

- **Episode Day 2, One:** [161]
  - **OBSERVATION, ≥ One:**
    - **General, One:**
      - **Responder**, discharge expected today if clinically stable last 12h, ≥ **One:** [162]
        - Abuse or neglect ruled out or alternative living arranged
        - Foreign body extracted without complications
        - Pain controlled or manageable [163]
        - Snakebite and, **All:**
          - Neurological stability [164]
          - No coagulopathy
          - No progression of symptoms
          - Peripheral circulation, sensation, and movement intact
      - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
        - Abuse or neglect and alternative placement required [4]
        - Foreign body and, **One:**
          - Post extraction monitoring at least every 4h [7, 8]
          - Repeat imaging ≤ 24h [165]
        - Pain, uncontrolled and, **One:**
          - Analgesic ≥ 2x/24h
          - Transition to PO analgesic ≤ 24h
      - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
        - For current trauma related condition use Episode Day 2 at a higher level of care
        - For a condition other than trauma use appropriate subset (Episode Day 1) based on clinical findings
        - Refer for secondary review
    - **Cardiovascular, One:**
      - **Responder**, discharge expected today if clinically stable last 12h, **One:** [162]
        - ECG normal or at baseline [17]
      - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
        - For current trauma related condition use Episode Day 2 at a higher level of care
        - For a condition other than trauma use appropriate subset (Episode Day 1) based on clinical findings
        - Refer for secondary review
    - **Gastrointestinal, Genitourinary, or biliary, One:**
      - **Responder**, discharge expected today if clinically stable last 12h, ≥ **One:** [162]
        - Abdominal trauma and pregnancy and, **Both:**
          - Fetal heart rate stable
          - No contractions
        - Organ injury ruled out
      - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
        - For current trauma related condition use Episode Day 2 at a higher level of care
        - For a condition other than trauma use appropriate subset (Episode Day 1) based on clinical findings
        - Refer for secondary review
    - **Musculoskeletal, One:**
      - **Responder**, discharge expected today if clinically stable last 12h, ≥ **One:** [162]

- Chest wall fracture and O_2 Sat within acceptable limits [166]
- Fracture, subluxation, or dislocation and cleared for discharge
- **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
  - Functional impairment (new) requiring rehabilitation initiation ≤ 24h and, ≥ **One:** [167, 168]
    - PT evaluation and training
    - OT evaluation and training
    - SLP evaluation and training
    - Cognitive evaluation and retraining
    - Swallowing evaluation and retraining
- **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
  - For current trauma related condition use Episode Day 2 at a higher level of care
  - For a condition other than trauma use appropriate subset (Episode Day 1) based on clinical findings
  - Refer for secondary review
- **Neurological, One:**
  - **Responder**, discharge expected today if clinically stable last 12h, **One:** [162]
    - Head trauma and neurological stability [164]
  - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
    - For current trauma related condition use Episode Day 2 at a higher level of care
    - For a condition other than trauma use appropriate subset (Episode Day 1) based on clinical findings
    - Refer for secondary review
- **Respiratory, One:**
  - **Responder**, discharge expected today if clinically stable last 12h, ≥ **One:** [162]
    - Chest tube and, **One:**
      - Heimlich valve
      - Removed and no reaccumulation of air confirmed by CXR
    - Hemothorax, pneumothorax or pneumomediastinum, **Both:** [39]
      - Resolving or resolved on CXR
      - O_2 sat within acceptable limits [166]
  - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
    - O_2 sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen [17, 85]
  - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
    - For current trauma related condition use Episode Day 2 at a higher level of care
    - For a condition other than trauma use appropriate subset (Episode Day 1) based on clinical findings
    - Refer for secondary review
- **Skin, One:**
  - **Responder**, discharge expected today if clinically stable last 12h, **One:** [162]
    - Patient or caregiver able to adhere to treatment regimen
    - Outpatient burn services coordinated
  - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
    - Burn and, **Both:** [46]
      - Patient or caregiver unable to adhere to treatment regimen
      - Outpatient services unavailable [47]
  - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
    - For other trauma related condition use Episode Day 2 at a higher level of care
    - For a condition other than trauma use appropriate subset (Episode Day 1) based on clinical findings
    - Refer for secondary review
- **ACUTE, ≥ One:**
  - **General, ≥ One:**
    - AKI and, **Both:** [169]

- Finding, ≥ **One:**
  - Urine output < 0.5 mL/kg/h
  - Creatinine, **One:**
    - ≥ 1.5x baseline and chronic kidney disease (170)
    - ≥ 1.5x ULN (171)
  - GFR > 25%(0.25) decrease from baseline (17)
  - Dialysis initiated
- Intervention, ≥ **One:**
  - Volume expander ≤ 2d (172, 173)
  - Diuretic ≥ 2x/24h, ≤ 3d
  - Medication adjustment or discontinuation ≤ 3d and, ≥ **One:** (174)
    - Diuretic (includes PO)
    - Nephrotoxic agent (includes PO) (175)
  - Dialysis (initial) and ≤ 7d since initiation (176)
  - Dialysis discontinued requiring lab monitoring ≤ 2d since discontinuation (177)

- Bleeding and, **Both:**
  - Finding, ≥ **One:**
    - Hct or Hb decreased within last 24h
    - Heart rate 100-120/min, sustained (178, 179)
  - Intervention, ≥ **One:**
    - Blood product transfusion
    - Hct or Hb monitoring at least 2x/24h, ≤ 2d
- Frailty and, ≥ **One:**
  - Rehabilitation evaluation scheduled or performed within 24h and, ≥ **One:** (180)
    - Analgesic (includes PO) ≥ 2x/24h
    - Continuous cardiac monitoring (excludes Holter)
    - Inadequate oral intake and IV fluid (75, 181)
    - Neurological assessment at least 4x/24h (12)
    - Oximetry monitoring every 4h
  - Specialty consultation ≤ 24h (182, 183)
- Post Critical care ≤ 24h

- **Cardiovascular or peripheral vascular, One:**
  - **Responder,** discharge expected today if clinically stable last 12h, **One:** (162)
    - Compartment syndrome or limb ischemia and, **Both:**
      - Ruled out or resolved
      - Peripheral circulation, sensation, and movement intact
  - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
    - Vascular compromise and neurovascular assessment at least 6x/24h, ≤ 2d (13)
- **Eye, One:**
  - **Responder,** discharge expected today if clinically stable last 12h, **All:** (162)
    - Ocular foreign body removed or open globe repaired
    - No signs of infection
    - Pain controlled or manageable (163)
  - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
    - Eye injury and, ≥ **One:**
      - Analgesic ≥ 4x/24h
      - Endophthalmitis and anti-infective ≤ 7d since initiation (184)
      - Post-op anti-infective ≤ 3d since initiation (185)
- **Gastrointestinal, Genitourinary, or biliary, One:**
  - **Responder,** discharge expected today if clinically stable last 12h, **One:** (162)
    - Organ injury and, ≥ **One:**
      - Lab values within acceptable limits (166)
      - Tolerating PO (186)

- Voiding without difficulty
- **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
  - Organ injury, actual and, ≥ **One:** [67, 68, 153]
    - Blood product transfusion
    - Hct or Hb monitoring at least 4x/24h
    - NPO and IV fluid [75, 154]
    - Suprapubic catheter, nephrostomy tube, or ureteral stent placed within 24h [155]
- **Musculoskeletal, ≥ One:**
  - Rhabdomyolysis or crush syndrome **(use Rhabdomyolysis or Crush Syndrome subset)**
  - Chest wall fracture and, ≥ **One:** [26]
    - Analgesic, continuous
    - Analgesic, regional ≤ 3d [70]
    - Respiratory impairment or high risk, **Both:** [71]
      - Finding, ≥ **One:**
        - Age > 65 yrs and > 2 rib fractures [72]
        - Dyspnea [73]
        - Incentive spirometry < 1000 mL
        - O_2 sat ≤ 91%(0.91) and < baseline [17]
      - Intervention, ≥ **One:**
        - Supplemental oxygen [85]
        - Respiratory interventions at least 6x/24h, ≤ 4d [156]
  - Facial fracture and, ≥ **One:**
    - CSF leak and neurological assessment at least 4x/24h, ≤ 7d [12, 74]
    - Inability to take PO and IV fluid [75]
    - Risk for airway compromise and oximetry monitoring every 4h, ≤ 2d [76]
  - Fracture, subluxation, or dislocation and, ≥ **One:** [187]
    - Analgesic ≥ 4x/24h or continuous
    - Analgesic, regional ≤ 3d [78]
    - Traction ≤ 4d (includes OR placement) [80]
- **Neurological, One:**
  - **Responder,** discharge expected today if clinically stable last 12h, ≥ **One:** [162]
    - CSF leak resolved
    - Head injury and neurological stability [164]
  - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
    - Head trauma and, **Both:**
      - Finding, ≥ **One:**
        - CSF leak [157]
        - GCS 9-14 or < baseline [17, 32, 33, 82]
      - Neurological assessment at least 4x/24h [12]
- **Respiratory, One:**
  - **Responder,** discharge expected today if clinically stable last 12h, **One:** [162]
    - Drowning and, ≥ **One:**
      - No evidence of pulmonary edema
      - O_2 sat within acceptable limits [166]
  - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - Chest tube and, ≥ **One:** [188]
      - Suction, continuous
      - Drainage ≥ 250 mL/24h
      - Repositioning within last 24h [189]
      - Water seal within last 24h
      - Video-assisted thoracoscopic surgery performed within last 24h
    - O_2 sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen [17, 85]
- **Skin, ≥ One:**

- Burn therapy, ≥ **One:** [190]
  - Complex wound care [92, 93]
  - Hydration and nutrition, ≥ **One:** [94]
    - Enteral feeds
    - IV fluid
    - TPN
  - Medication, ≥ **One:**
    - Analgesic ≥ 4x/24h or continuous [95]
    - Sedative ≥ 3x/24h [96]
  - Neurovascular assessment at least 6x/24h [13, 97]
  - O_2 sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen [17, 85]
- Complex wound care, ≥ **One:** [191]
  - Wound care ≥ 3x/24h, ≤ 5d and, ≥ **One:**
    - Requiring parenteral analgesic
    - ≥ 15 min in duration
  - Wound management teaching ≤ 24h [192]

INTERMEDIATE, ≥ One:
- Accidental hypothermia and active rewarming [99, 100, 158]
- Arrhythmia, ≥ **One:**
  - IV antiarrhythmic and hemodynamically stable
  - Antiarrhythmic conversion from IV to PO, **One:**
    - No proarrhythmic risk ≤ 24h
    - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) ≤ 3d [193]
- Chest trauma or electrical injury and continuous cardiac monitoring (excludes Holter) [14, 194]
- Complex wound care ≤ 5d and, **Both:** [191]
  - ≥ 3x/24h
  - ≥ 30 min in duration
- Decompression sickness and, **One:** [102, 103]
  - Oxygen 100%(1.00) [85]
  - Hyperbaric oxygen therapy (HBOT)
- External ventricular or lumbar drainage (includes OR placement) and neurological assessment every 3-4h [12]
- HFNC and respiratory monitoring every 3-4h [107, 108]
- High risk trauma and neurological assessment every 3-4h [12]
- Neurological impairment, **Both:** [195]
  - Acute onset or deterioration
  - Neurological assessment every 3-4h, ≤ 2d [12]
- NIPPV [109]
- Oxygen ≥ 40%(0.40) [85]
- Post permanent pacemaker or ICD placement within last 24h

CRITICAL, ≥ One:
- Rhabdomyolysis or crush syndrome (**use Rhabdomyolysis or Crush Syndrome subset**)
- Abdominal trauma and, **One:**
  - Angioembolization ≤ 24h [112]
  - Post angioembolization ≤ 24h
- Accidental hypothermia and, **Both:** [99, 100]
  - Hemodynamic instability [113]
  - Active rewarming [158]
- AKI and, **Both:** [169]
  - Finding, ≥ **One:**
    - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One:**
      - Loss of P wave
      - Multifocal PVCs

- Neuromuscular deficit [196]
- Widening QRS
- Sodium, **One:**
  - < 120 mEq/L(120 mmol/L) and volume overload
  - > 160 mEq/L(160 mmol/L)
- Intervention, ≥ **One:**
  - Calcium chloride or gluconate
  - IV fluid resuscitation ≤ 2d [133]
  - Diuretic, continuous
  - Dialysis, **One:**
    - Hemodialysis ≤ 7d since initiation
    - Peritoneal dialysis ≤ 7d since initiation
- Aortic injury and, ≥ **One:** [114]
  - Antihypertensive, continuous
  - Endovascular repair ≤ 24h
  - Post endovascular repair ≤ 24h
  - Hct or Hb monitoring at least 4x/24h, ≤ 2d
- Arrhythmia, ≥ **One:**
  - Urgent electrical cardioversion performed within last 24h
  - Defibrillation performed within last 24h
  - Temporary pacemaker insertion performed within last 2d
  - Antiarrhythmic and, **Both:**
    - Arrhythmia, ≥ **One:**
      - Recurrent arrhythmia [197]
      - Refractory arrhythmia [197]
    - Symptomatic, ≥ **One:**
      - Chest pain
      - Heart failure and, ≥ **One:**
        - Arterial Po_2 < 56 mmHg(7.4 kPa)
        - O_2 sat < 89%(0.89) and < baseline [17]
      - Hemodynamic instability, ≥ **One:**
        - Heart rate > 120/min, sustained [179]
        - Systolic blood pressure, ≥ **One:**
          - < 90 mmHg
          - < 110 mmHg with chronic hypertension
          - > 30 mmHg decrease from baseline [17]
          - Labile [198]
        - MAP < 65 mmHg
- Burn therapy, ≥ **One:** [190]
  - Anxiolytic, continuous [116]
  - IV fluid resuscitation and urine output monitoring every 1-2h [117]
  - Neurovascular assessment every 1-2h [13, 97]
  - Wound care requiring procedural sedation [92, 118, 119]
- Cardiac arrest and post resuscitation care ≤ 2d [120]
- Cerebral edema and, ≥ **One:**
  - ICP monitoring (includes OR placement)
  - Hyperosmolar therapy, ≥ **One:** [122]
    - Mannitol
    - Hypertonic saline ≥ 3%(0.03) solution
- Complex wound care and, ≥ **One:**
  - Every 1-2h
  - Requiring procedural sedation [118]
- CRRT [199]

- DIC and blood product transfusion [123]
- ECMO or ECLS [124]
- Emergent tracheostomy or cricothyrotomy ≤ 24h [130]
- External ventricular or lumbar drainage (includes OR placement) and neurological assessment every 1-2h [12]
- Flexible bronchoscopy ≥ 2x/24h [131]
- Hemodynamic instability persisting after ≥ 1L of IVF resuscitation or ≥ 1 unit blood product transfusion and, **Both:** [132]
  - Finding, ≥ **One:**
    - Heart rate > 120/min
    - Systolic blood pressure < 100 mmHg and < baseline [17]
    - MAP < 65 mmHg
  - Intervention, ≥ **One:**
    - Blood product transfusion
    - IV fluid resuscitation ≥ 1L, ≤ 24h [133]
- Hemodynamic monitoring, invasive [134]
- HFNC and respiratory monitoring every 1-2h [107, 108]
- IABP [135]
- Mechanical ventilation
- Nebulizer or MDI treatment, **One:** [136]
  - Every 1-2h
  - Continuous
- Neurological impairment, **Both:** [195]
  - Acute onset or deterioration
  - Neurological assessment every 1-2h, ≤ 2d [12]
- NIPPV and, ≥ **One:** [109]
  - FiO_2 > 50%(0.50)
  - Respiratory intervention every 1-2h [200]
- Organ injury, **Both:** [142]
  - Finding, ≥ **One:**
    - Grade ≥ III liver, pancreas, or spleen injury [143]
    - Grade ≥ IV kidney injury [144]
  - Intervention, ≥ **One:** [153]
    - Hct or Hb monitoring at least 4x/24h
    - Blood product transfusion
- Pelvic fracture and, ≥ **One:**
  - Angioembolization ≤ 24h [145]
  - Post angioembolization ≤ 24h
  - Post pelvic external fixator
- Snakebite and, **One:**
  - Antivenom and, ≥ **One:** [149]
    - Neurological assessment every 1-2h [12]
    - Neurovascular assessment every 1-2h [13]
  - Post antivenom ≤ 24h and, ≥ **One:** [201]
    - Neurological assessment every 1-2h [12]
    - Neurovascular assessment every 1-2h [13]
- Vascular compromise, **Both:**
  - Finding, ≥ **One:**
    - Compartment syndrome, actual or suspected [63]
    - Limb ischemia, actual or suspected
  - Intervention, ≥ **One:**
    - Endovascular revascularization ≤ 24h [150]
    - Post endovascular revascularization ≤ 24h

- Neurovascular assessment every 1-2h [13]
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

- **Episode Day 3-X, One:** [202]
  - **OBSERVATION, ≥ One:**
    - **General, One:**
      - **Responder**, discharge expected today if clinically stable last 12h, ≥ **One:** [162]
        - Abuse or neglect and alternative living arranged
        - Foreign body extracted without complications
        - Pain controlled or manageable [163]
      - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
        - For current trauma related condition use Episode Day 3-X at a higher level of care
        - For a condition other than trauma use appropriate subset (Episode Day 1) based on clinical findings
        - Refer for secondary review
    - **Musculoskeletal, One:**
      - **Responder**, discharge expected today if clinically stable last 12h, **One:** [162]
        - Functional impairment and, **One:** [167]
          - Discharge to home, **One:** [203]
            - Patient or caregiver able to safely manage impairment
            - Home care services arranged
          - Discharge to inpatient facility for rehabilitative services
      - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
        - For current trauma related condition use Episode Day 3-X at a higher level of care
        - For a condition other than trauma use appropriate subset (Episode Day 1) based on clinical findings
        - Refer for secondary review
    - **Respiratory, One:**
      - **Responder**, discharge expected today if clinically stable last 12h, **One:** [162]
        - O_2 sat within acceptable limits [166]
      - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
        - For current trauma related condition use Episode Day 3-X at a higher level of care
        - For a condition other than trauma use appropriate subset (Episode Day 1) based on clinical findings
        - Refer for secondary review
    - **Skin, One:**
      - **Responder**, discharge expected today if clinically stable last 12h, **One:** [162]
        - Patient or caregiver able to adhere to treatment regimen
        - Outpatient burn services coordinated
      - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
        - For other trauma related condition use Episode Day 3-X at a higher level of care
        - For a condition other than trauma use appropriate subset (Episode Day 1) based on clinical findings
        - Refer for secondary review
  - **ACUTE, ≥ One:**
    - **General, One:**
      - **Responder**, discharge expected today if clinically stable last 12h, ≥ **One:** [162]
        - Afebrile
        - AKI and, ≥ **One:**
          - Kidney function improving or within acceptable limits [166]
          - Outpatient dialysis regimen established
        - Bleeding controlled or resolved
        - Frailty or functional impairment and, **One:** [167]
          - Discharge to home and, **One:** [203]

- Patient or caregiver able to safely manage impairment
- Home care services arranged
- Discharge to inpatient facility for rehabilitative services
- Pain controlled or manageable [163]
- Tolerating PO [186]

- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
  - AKI and, **Both:** [169]
    - Finding, ≥ **One:**
      - Urine output < 0.5 mL/kg/h
      - Creatinine, **One:**
        - ≥ 1.5x baseline and chronic kidney disease [170]
        - ≥ 1.5x ULN [171]
      - GFR > 25%(0.25) decrease from baseline [17]
      - Dialysis initiated
    - Intervention, ≥ **One:**
      - Volume expander ≤ 2d [172, 173]
      - Diuretic ≥ 2x/24h, ≤ 3d
      - Medication adjustment or discontinuation ≤ 3d and, ≥ **One:** [174]
        - Diuretic (includes PO)
        - Nephrotoxic agent (includes PO) [175]
      - Dialysis (initial) and ≤ 7d since initiation [176]
      - Dialysis discontinued requiring lab monitoring ≤ 2d since discontinuation [177]
  - Bleeding and, **Both:**
    - Finding, ≥ **One:**
      - Hct or Hb decreased within last 24h
      - Heart rate 100-120/min, sustained [178, 179]
    - Intervention, ≥ **One:**
      - Blood product transfusion
      - Hct or Hb monitoring at least 2x/24h, ≤ 2d
  - Frailty and specialty consultation ≤ 24h [182, 183]
  - Functional impairment (new) requiring rehabilitation initiation ≤ 24h and, ≥ **One:** [167, 168]
    - PT evaluation and training
    - OT evaluation and training
    - ST evaluation and training
    - Cognitive evaluation and retraining
    - Swallowing evaluation and retraining
  - Inadequate oral intake and, ≥ **One:** [181]
    - IV fluid ≤ 2d [75]
    - Parenteral nutrition ≤ 2d
    - Initiation or adjustment of PO or tube feeding ≤ 2d [204]
  - Infection, actual or suspected and, ≥ **One:**
    - Anti-infective ≤ 7d since initiation [205]
    - Culture pending ≤ 2d
    - Drug-induced fever suspected and precipitating drug discontinued ≤ 3d [206]
    - Imaging study ≤ 24h
  - Pain, uncontrolled and, **One:**
    - Analgesic ≥ 3x/24h or continuous
    - Analgesic, regional ≤ 3d
    - Requiring change in medication, dose, or frequency ≤ 2d [174, 207]
    - Transition to PO analgesic ≤ 24h
  - Post Critical care ≤ 24h
- **Non-responder**, not clinically stable for discharge, **One:**
  - Does not meet partial responder criteria, **One:**

- Use Extended Stay subset
- Refer for secondary review

- **Cardiovascular or peripheral vascular, One:**
  - **Responder,** discharge expected today if clinically stable last 12h, ≥ **One:** (162)
    - Aortic injury and, **Both:**
      - Blood pressure within acceptable limits (166)
      - Lab values within acceptable limits (166)
    - Compartment syndrome or limb ischemia and, **Both:**
      - Ruled out or resolved
      - Peripheral circulation, sensation, and movement intact
  - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - Aortic injury and, ≥ **One:**
      - Blood pressure management, **Both:**
        - Blood pressure monitoring at least 3x/24h
        - Requiring change in antihypertensive medication, dose, or frequency (includes PO) ≤ 2d (174)
      - Repeat imaging ≤ 24h (208)
    - Vascular compromise and neurovascular assessment at least 6x/24h, ≤ 2d (13)
  - **Non-responder**, not clinically stable for discharge, **One:**
    - Does not meet partial responder criteria, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review
- **Eye, One:**
  - **Responder,** discharge expected today if clinically stable last 12h, ≥ **One:** (162)
    - Pain controlled or manageable (163)
    - Anti-infective arranged or completed
  - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
    - Eye injury and, ≥ **One:** (209)
      - Endophthalmitis and anti-infective ≤ 7d since initiation (184)
      - Post-op anti-infective ≤ 3d since initiation (185)
  - **Non-responder**, not clinically stable for discharge, **One:**
    - Does not meet partial responder criteria, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review
- **Gastrointestinal, Genitourinary or biliary, One:**
  - **Responder**, discharge expected today if clinically stable last 12h, **One:** (162)
    - Organ injury and, ≥ **One:**
      - Lab values within acceptable limits (166)
      - Tolerating PO (186)
      - Voiding without difficulty
  - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
    - Organ injury and, **Both:** (67, 68)
      - Finding, **One:** (143, 144)
        - Grade II, ≤ 3d since injury
        - Grade III, ≤ 4d since injury
        - Grade IV, ≤ 5d since injury
        - Grade V, ≤ 6d since injury
      - Intervention, ≥ **One:** (153)
        - Hct or Hb monitoring at least 2x/24h
        - Blood product transfusion
  - **Non-responder**, not clinically stable for discharge, **One:**
    - Does not meet partial responder criteria, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review

- **Musculoskeletal, One:**
  - **Responder,** discharge expected today if clinically stable last 12h, ≥ **One:** (162)
    - Chest wall fracture and, ≥ **One:**
      - Dyspnea resolved
      - O_2 sat within acceptable limits (166)
      - Pain controlled or manageable (163)
    - Facial fracture and, ≥ **One:**
      - Airway patent
      - CSF leak resolved
      - Tolerating PO (186)
    - Fracture, subluxation, or dislocation stabilized
  - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:** (209)
    - Rhabdomyolysis or crush syndrome (**use Rhabdomyolysis or Crush Syndrome subset**)
    - Chest wall fracture and, ≥ **One:** (26)
      - O_2 sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen (17, 85)
      - Respiratory interventions at least 6x/24h, ≤ 4d (156)
    - CSF leak and neurological assessment at least 4x/24h, ≤ 7d (12, 74)
    - Pelvic external fixator in place ≤ 3d (210)
    - Traction ≤ 4d (includes OR placement) (80)
  - **Non-responder**, not clinically stable for discharge, **One:**
    - Does not meet partial responder criteria, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review
- **Neurological, One:**
  - **Responder,** discharge expected today if clinically stable last 12h, ≥ **One:** (162)
    - CSF leak resolved
    - Neurological stability (164)
  - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - CSF leak and neurological assessment at least 4x/24h, ≤ 5d (12, 157)
    - Neurological impairment and, **Both:** (195)
      - Acute onset or deterioration
      - Neurological assessment at least 4x/24h, ≤ 2d (12)
  - **Non-responder**, not clinically stable for discharge, **One:**
    - Does not meet partial responder criteria, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review
- **Respiratory, One:**
  - **Responder,** discharge expected today if clinically stable last 12h, ≥ **One:** (162)
    - O_2 sat within acceptable limits (166)
    - Chest tube and, **One:** (188)
      - Heimlich valve
      - Removed and no reaccumulation of air or fluid confirmed by CXR
    - Decompression sickness and, **Both:**
      - Dyspnea absent or resolved
      - Neurological stability (164)
  - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - Chest tube and, ≥ **One:** (188)
      - Suction, continuous
      - Drainage ≥ 250 mL/24h
      - Repositioning within last 24h (189)
      - Water seal within last 24h
      - Video-assisted thoracoscopic surgery performed within last 24h
    - Hyperbaric oxygen therapy (HBOT) ≤ 3d since initiation (102, 103)

- Respiratory impairment and, **Both:**
  - Finding, ≥ **One:**
    - Inability to clear secretions ≤ 24h
    - O_2 sat ≤ 91%(0.91) and < baseline [17]
  - Intervention, ≥ **One:**
    - Requiring supplemental oxygen [85]
    - Respiratory interventions at least 6x/24h [211]
- **Non-responder**, not clinically stable for discharge, **One:**
  - Does not meet partial responder criteria, **One:**
    - **Use Extended Stay subset**
    - Refer for secondary review
- **Skin, One:**
  - **Responder**, discharge expected today if clinically stable last 12h, **Both:** [162]
    - Pain controlled or manageable [163]
    - Clinical stability, ≥ **One:**
      - Burn healing or manageable and, ≥ **One:**
        - O_2 sat within acceptable limits [166]
        - Peripheral circulation, sensation, and movement intact
        - Tolerating PO or nutritional route established [186, 212]
      - Wound healing or manageable
  - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - Burn therapy, ≥ **One:** [190]
      - Complex wound care [92, 93]
      - O_2 sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen [17, 85]
      - Pain, uncontrolled and, **One:**
        - Analgesic ≥ 3x/24h or continuous [95]
        - Requiring change in medication, dose, or frequency ≤ 2d [174, 207]
        - Transition to PO analgesic ≤ 24h
      - Sedative ≥ 3x/24h [96]
    - Complex wound care, ≥ **One:** [191]
      - Wound care ≥ 3x/24h, ≤ 5d and, ≥ **One:**
        - Requiring parenteral analgesic
        - ≥ 15 min in duration
      - Wound management teaching ≤ 24h [192]
  - **Non-responder**, not clinically stable for discharge, **One:**
    - Does not meet partial responder criteria, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review
- **INTERMEDIATE, One:**
  - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - Arrhythmia, ≥ **One:**
      - IV antiarrhythmic and hemodynamically stable
      - Antiarrhythmic conversion from IV to PO, **One:**
        - No proarrhythmic risk ≤ 24h
        - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) ≤ 3d [193]
    - Complex wound care ≤ 5d and, **Both:** [191]
      - ≥ 3x/24h
      - ≥ 30 min in duration
    - External ventricular or lumbar drainage (includes OR placement) and neurological assessment every 3-4h [12]
    - HFNC and respiratory monitoring every 3-4h [107, 108]
    - Neurological impairment, **Both:** [195]
      - Acute onset or deterioration

        - Neurological assessment every 3-4h, ≤ 2d (12)
      - NIPPV (109)
      - Oxygen ≥ 40%(0.40) (85)
      - Post permanent pacemaker or ICD placement within last 24h
    - **Non-responder**, not clinically stable for discharge, **One:**
      - Does not meet partial responder criteria or at the Acute level of care, **One:**
        - **Use Extended Stay subset**
        - Refer for secondary review
- **CRITICAL, One:**
  - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - Rhabdomyolysis or crush syndrome (**use Rhabdomyolysis or Crush Syndrome subset**)
    - Abdominal trauma and, **One:**
      - Angioembolization ≤ 24h (112)
      - Post angioembolization ≤ 24h
    - AKI and, **Both:** (169)
      - Finding, ≥ **One:**
        - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One:**
          - Loss of P wave
          - Multifocal PVCs
          - Neuromuscular deficit (196)
          - Widening QRS
        - Sodium, **One:**
          - < 120 mEq/L(120 mmol/L) and volume overload
          - > 160 mEq/L(160 mmol/L)
      - Intervention, ≥ **One:**
        - Calcium chloride or gluconate
        - IV fluid resuscitation ≤ 2d (133)
        - Diuretic, continuous
        - Dialysis, **One:**
          - Hemodialysis ≤ 7d since initiation
          - Peritoneal dialysis ≤ 7d since initiation
    - Aortic injury and, ≥ **One:** (114)
      - Antihypertensive, continuous
      - Endovascular repair ≤ 24h
      - Post endovascular repair ≤ 24h
      - Hct or Hb monitoring at least 4x/24h, ≤ 2d
    - Arrhythmia, ≥ **One:**
      - Urgent electrical cardioversion performed within last 24h
      - Defibrillation performed within last 24h
      - Temporary pacemaker insertion performed within last 2d
      - Antiarrhythmic and, **Both:**
        - Arrhythmia, ≥ **One:**
          - Recurrent arrhythmia (197)
          - Refractory arrhythmia (197)
        - Symptomatic, ≥ **One:**
          - Chest pain
          - Heart failure and, ≥ **One:**
            - Arterial Po_2 < 56 mmHg(7.4 kPa)
            - O_2 sat < 89%(0.89) and < baseline (17)
          - Hemodynamic instability, ≥ **One:**
            - Heart rate > 120/min, sustained (179)
            - Systolic blood pressure, ≥ **One:**
              - < 90 mmHg

- < 110 mmHg with chronic hypertension
- > 30 mmHg decrease from baseline [17]
- Labile [198]
- MAP < 65 mmHg

⊖ Burn therapy, ≥ **One:** [190]
- Anxiolytic, continuous [116]
- IV fluid resuscitation and urine output monitoring every 1-2h [117]
- Neurovascular assessment every 1-2h [13, 97]
- Wound care requiring procedural sedation [92, 118, 119]

- Cardiac arrest and post resuscitation care ≤ 2d [120]

⊖ Cerebral edema and, ≥ **One:**
- ICP monitoring (includes OR placement)
- ⊖ Hyperosmolar therapy, ≥ **One:** [122]
  - Mannitol
  - Hypertonic saline ≥ 3%(0.03) solution

⊖ Complex wound care and, ≥ **One:**
- Every 1-2h
- Requiring procedural sedation [118]

- CRRT [199]
- DIC and blood product transfusion [123]
- ECMO or ECLS [124]
- Emergent tracheostomy or cricothyrotomy ≤ 24h [130]
- External ventricular or lumbar drainage (includes OR placement) and neurological assessment every 1-2h [12]
- Flexible bronchoscopy ≥ 2x/24h [131]

⊖ Hemodynamic instability persisting after ≥ 1L of IVF resuscitation or ≥ 1 unit blood product transfusion and, **Both:** [132]
- ⊖ Finding, ≥ **One:**
  - Heart rate > 120/min
  - Systolic blood pressure < 100 mmHg and < baseline [17]
  - MAP < 65 mmHg
- ⊖ Intervention, ≥ **One:**
  - Blood product transfusion
  - IV fluid resuscitation ≥ 1L, ≤ 24h [133]

- Hemodynamic monitoring, invasive [134]
- HFNC and respiratory monitoring every 1-2h [107, 108]
- IABP [135]
- Mechanical ventilation

⊖ Nebulizer or MDI treatment, **One:** [136]
- Every 1-2h
- Continuous

⊖ Neurological impairment, **Both:** [195]
- Acute onset or deterioration
- Neurological assessment every 1-2h, ≤ 2d [12]

⊖ NIPPV and, ≥ **One:** [109]
- FiO_2 > 50%(0.50)
- Respiratory intervention every 1-2h [200]

- Pelvic fracture and post angioembolization ≤ 24h

⊖ Snakebite and post antivenom ≤ 24h and, ≥ **One:** [201]
- Neurological assessment every 1-2h [12]
- Neurovascular assessment every 1-2h [13]

⊖ Vascular compromise, **Both:**
- ⊖ Finding, ≥ **One:**

- Compartment syndrome, actual or suspected [63]
- Limb ischemia, actual or suspected
- Intervention, ≥ **One:**
  - Endovascular revascularization ≤ 24h [150]
  - Post endovascular revascularization ≤ 24h
  - Neurovascular assessment every 1-2h [13]
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Non-responder**, not clinically stable for discharge, **One:**
  - Does not meet partial responder criteria or at the Acute or Intermediate level of care, **One:**
    - **Use Extended Stay subset**
    - Refer for secondary review

Discharge, Level of Care, One: [213]

- **HOME, Both:**
  - Level of care appropriateness, **Both:**
    - Home environment safe and accessible [214]
    - Patient or caregiver demonstrates ability to manage care
  - Discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions [215]
    - Medication reconciliation [216]
    - Follow-up care planned
    - Identify and address transportation needs
- **HOME CARE, Both:**
  - Level of care appropriateness, **All:**
    - Home environment safe and accessible [214]
    - Patient or caregiver willing and able to learn care
    - Outpatient management contraindicated or unavailable [47, 217]
    - Skilled services, ≥ **One:** [218]
      - Skilled nursing [219]
      - Skilled therapy, PT, OT, or ST
      - Behavioral health [220]
  - Discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions [215]
    - Medication reconciliation [216]
    - Follow-up care planned
    - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
  - Level of care appropriateness, **One:**
    - Support systems available [221]
  - Skilled treatment, ≥ **One:**
    - Clinical assessment [222]
    - Individual, group, or family counseling
    - Psychiatric, substance, or medication evaluation
- **SKILLED MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 1-2x/24h [219]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [223, 224]
        - Functional impairments requiring at least supervision [225]
        - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk

- Complete prior to facility transfer, **All:**
  - Medication reconciliation [216]
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 4h/24h [219]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [223, 224]
        - ≥ 2 functional impairments requiring at least minimum assistance [225]
        - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [216]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE REHAB, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Treatment precluded in a lower level
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [223, 224]
      - ≥ 2 functional impairments requiring at least minimum assistance [225]
      - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [216]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **ACUTE REHAB, Both:**
  - Level of care appropriateness, **Both:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [223, 224]
      - ≥ 2 functional impairments requiring at least minimum assistance [225]
      - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines [226]
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [216]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
  - Level of care appropriateness, **Both:**
    - Treatment precluded in a lower level
    - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
      - Primary condition or illness and treatment of active comorbid condition(s) [227]
      - Ventilator dependent ≥ 6h/d and weaning planned [228, 229]
      - Acute and comorbid conditions requiring prolonged hospitalization [230]
  - Complete prior to facility transfer, **All:**

- Medication reconciliation [216]
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
052 SPINAL DISORDERS AND INJURIES WITH CC/MCC	**n/a**	**4.6**	**CMS GMLOS**
053 SPINAL DISORDERS AND INJURIES WITHOUT CC/MCC	**n/a**	**2.8**	**CMS GMLOS**
073 CRANIAL AND PERIPHERAL NERVE DISORDERS WITH MCC	**n/a**	**3.9**	**CMS GMLOS**
074 CRANIAL AND PERIPHERAL NERVE DISORDERS WITHOUT MCC	**n/a**	**2.9**	**CMS GMLOS**
082 TRAUMATIC STUPOR AND COMA >1 HOUR WITH MCC	**n/a**	**4.5**	**CMS GMLOS**
083 TRAUMATIC STUPOR AND COMA >1 HOUR WITH CC	**n/a**	**3.3**	**CMS GMLOS**
084 TRAUMATIC STUPOR AND COMA >1 HOUR WITHOUT CC/MCC	**n/a**	**2.1**	**CMS GMLOS**
088 CONCUSSION WITH MCC	**n/a**	**3.9**	**CMS GMLOS**
089 CONCUSSION WITH CC	**n/a**	**2.5**	**CMS GMLOS**
090 CONCUSSION WITHOUT CC/MCC	**n/a**	**1.9**	**CMS GMLOS**
091 OTHER DISORDERS OF NERVOUS SYSTEM WITH MCC	**n/a**	**4.5**	**CMS GMLOS**
092 OTHER DISORDERS OF NERVOUS SYSTEM WITH CC	**n/a**	**3.1**	**CMS GMLOS**
093 OTHER DISORDERS OF NERVOUS SYSTEM WITHOUT CC/MCC	**n/a**	**2.2**	**CMS GMLOS**
124 OTHER DISORDERS OF THE EYE WITH MCC OR THROMBOLYTIC AGENT	**n/a**	**3.4**	**CMS GMLOS**
125 OTHER DISORDERS OF THE EYE WITHOUT MCC	**n/a**	**2.3**	**CMS GMLOS**
154 OTHER EAR, NOSE, MOUTH AND THROAT DIAGNOSES WITH MCC	**n/a**	**4.2**	**CMS GMLOS**
155 OTHER EAR, NOSE, MOUTH AND THROAT DIAGNOSES WITH CC	**n/a**	**2.9**	**CMS GMLOS**
156 OTHER EAR, NOSE, MOUTH AND THROAT DIAGNOSES WITHOUT CC/MCC	**n/a**	**2.1**	**CMS GMLOS**

Condition or Procedure	% Pd Obs	LOS (days)	Type
157 DENTAL AND ORAL DISEASES WITH MCC	n/a	4.7	CMS GMLOS
158 DENTAL AND ORAL DISEASES WITH CC	n/a	2.9	CMS GMLOS
159 DENTAL AND ORAL DISEASES WITHOUT CC/MCC	n/a	2	CMS GMLOS
183 MAJOR CHEST TRAUMA WITH MCC	n/a	4.5	CMS GMLOS
184 MAJOR CHEST TRAUMA WITH CC	n/a	3.1	CMS GMLOS
185 MAJOR CHEST TRAUMA WITHOUT CC/MCC	n/a	2.2	CMS GMLOS
199 PNEUMOTHORAX WITH MCC	n/a	5.1	CMS GMLOS
200 PNEUMOTHORAX WITH CC	n/a	3.2	CMS GMLOS
201 PNEUMOTHORAX WITHOUT CC/MCC	n/a	2.4	CMS GMLOS
205 OTHER RESPIRATORY SYSTEM DIAGNOSES WITH MCC	n/a	4.5	CMS GMLOS
206 OTHER RESPIRATORY SYSTEM DIAGNOSES WITHOUT MCC	n/a	2.5	CMS GMLOS
299 PERIPHERAL VASCULAR DISORDERS WITH MCC	n/a	4	CMS GMLOS
300 PERIPHERAL VASCULAR DISORDERS WITH CC	n/a	3.1	CMS GMLOS
301 PERIPHERAL VASCULAR DISORDERS WITHOUT CC/MCC	n/a	2.1	CMS GMLOS
368 MAJOR ESOPHAGEAL DISORDERS WITH MCC	n/a	4.2	CMS GMLOS
369 MAJOR ESOPHAGEAL DISORDERS WITH CC	n/a	3	CMS GMLOS
370 MAJOR ESOPHAGEAL DISORDERS WITHOUT CC/MCC	n/a	2.2	CMS GMLOS
393 OTHER DIGESTIVE SYSTEM DIAGNOSES WITH MCC	n/a	4.4	CMS GMLOS
394 OTHER DIGESTIVE SYSTEM DIAGNOSES WITH CC	n/a	3	CMS GMLOS
395 OTHER DIGESTIVE SYSTEM DIAGNOSES WITHOUT CC/MCC	n/a	2.1	CMS GMLOS
438 DISORDERS OF PANCREAS EXCEPT MALIGNANCY WITH MCC	n/a	4.7	CMS GMLOS
439 DISORDERS OF PANCREAS EXCEPT MALIGNANCY WITH CC	n/a	3.1	CMS GMLOS
440 DISORDERS OF PANCREAS EXCEPT MALIGNANCY WITHOUT CC/MCC	n/a	2.4	CMS GMLOS

Condition or Procedure	% Pd Obs	LOS (days)	Type
441 DISORDERS OF LIVER EXCEPT MALIGNANCY, CIRRHOSIS OR ALCOHOLIC HEPATITIS WITH MCC	n/a	4.8	CMS GMLOS
442 DISORDERS OF LIVER EXCEPT MALIGNANCY, CIRRHOSIS OR ALCOHOLIC HEPATITIS WITH CC	n/a	3.2	CMS GMLOS
443 DISORDERS OF LIVER EXCEPT MALIGNANCY, CIRRHOSIS OR ALCOHOLIC HEPATITIS WITHOUT CC/MCC	n/a	2.5	CMS GMLOS
533 FRACTURES OF FEMUR WITH MCC	n/a	4.6	CMS GMLOS
534 FRACTURES OF FEMUR WITHOUT MCC	n/a	2.8	CMS GMLOS
535 FRACTURES OF HIP AND PELVIS WITH MCC	n/a	3.9	CMS GMLOS
536 FRACTURES OF HIP AND PELVIS WITHOUT MCC	n/a	2.8	CMS GMLOS
551 MEDICAL BACK PROBLEMS WITH MCC	n/a	4.6	CMS GMLOS
552 MEDICAL BACK PROBLEMS WITHOUT MCC	n/a	2.9	CMS GMLOS
562 FRACTURE, SPRAIN, STRAIN AND DISLOCATION EXCEPT FEMUR, HIP, PELVIS AND THIGH WITH MCC	n/a	4.3	CMS GMLOS
563 FRACTURE, SPRAIN, STRAIN AND DISLOCATION EXCEPT FEMUR, HIP, PELVIS AND THIGH WITHOUT MCC	n/a	2.8	CMS GMLOS
564 OTHER MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE DIAGNOSES WITH MCC	n/a	5	CMS GMLOS
565 OTHER MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE DIAGNOSES WITH CC	n/a	3.3	CMS GMLOS
566 OTHER MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE DIAGNOSES WITHOUT CC/MCC	n/a	2.4	CMS GMLOS
604 TRAUMA TO THE SKIN, SUBCUTANEOUS TISSUE AND BREAST WITH MCC	n/a	3.9	CMS GMLOS
605 TRAUMA TO THE SKIN, SUBCUTANEOUS TISSUE AND BREAST WITHOUT MCC	n/a	2.6	CMS GMLOS
606 MINOR SKIN DISORDERS WITH MCC	n/a	4.7	CMS GMLOS
607 MINOR SKIN DISORDERS WITHOUT MCC	n/a	3	CMS GMLOS
913 TRAUMATIC INJURY WITH MCC	n/a	3.9	CMS GMLOS
914 TRAUMATIC INJURY WITHOUT MCC	n/a	2.5	CMS GMLOS
922 OTHER INJURY, POISONING AND TOXIC EFFECT DIAGNOSES WITH MCC	n/a	4.5	CMS GMLOS

Condition or Procedure	% Pd Obs	LOS (days)	Type
923 OTHER INJURY, POISONING AND TOXIC EFFECT DIAGNOSES WITHOUT MCC	n/a	2.9	CMS GMLOS
928 FULL THICKNESS BURN WITH SKIN GRAFT OR INHALATION INJURY WITH CC/MCC	n/a	12.6	CMS GMLOS
929 FULL THICKNESS BURN WITH SKIN GRAFT OR INHALATION INJURY WITHOUT CC/MCC	n/a	6	CMS GMLOS
933 EXTENSIVE BURNS OR FULL THICKNESS BURNS WITH MV >96 HOURS WITHOUT SKIN GRAFT	n/a	2.5	CMS GMLOS
935 NON-EXTENSIVE BURNS	n/a	3.7	CMS GMLOS
947 SIGNS AND SYMPTOMS WITH MCC	n/a	3.7	CMS GMLOS
948 SIGNS AND SYMPTOMS WITHOUT MCC	n/a	2.6	CMS GMLOS
951 OTHER FACTORS INFLUENCING HEALTH STATUS	n/a	1.8	CMS GMLOS
ABDOMINAL TRAUMA	52%	4.1	InterQual
BLEEDING, ACTUAL OR SUSPECTED	11%	5.2	InterQual
CHEST TRAUMA	31%	4.3	InterQual
COMPARTMENT SYNDROME	20%	6.1	InterQual
DISLOCATION OF MAJOR JOINT PROSTHESIS	23%	3.4	InterQual
EPIDURAL HEMATOMA	12%	5.2	InterQual
FACIAL FRACTURE	41%	3.5	InterQual
FALL	75%	4.4	InterQual
FLAIL CHEST	9%	7.2	InterQual
FOREIGN BODY	49%	3.7	InterQual
FRACTURE OF CERVICAL SPINE (EXCLUDES UNCOMPLICATED COMPRESSION FRACTURES)	23%	4.6	InterQual
FRACTURE OF FEMUR AND NOT A SURGICAL CANDIDATE	3%	5.3	InterQual
FRACTURE OF LUMBAR SPINE (EXCLUDES UNCOMPLICATED COMPRESSION FRACTURES)	37%	4.7	InterQual
FRACTURE OF PELVIS AND NOT A SURGICAL CANDIDATE	32%	6.3	InterQual
FRACTURE OF THORACIC SPINE (EXCLUDES UNCOMPLICATED COMPRESSION FRACTURES)	40%	4.7	InterQual

Condition or Procedure	% Pd Obs	LOS (days)	Type
HEAD TRAUMA	**30%**	**4.8**	**InterQual**
LIVER TRAUMA, GRADE II	**22%**	**4.6**	**InterQual**
LIVER TRAUMA, GRADE III	**22%**	**4.6**	**InterQual**
LIVER TRAUMA, GRADE IV	**22%**	**4.6**	**InterQual**
LIVER TRAUMA, GRADE V	**22%**	**4.6**	**InterQual**
LUNG CONTUSION	**25%**	**3.8**	**InterQual**
OCULAR FOREIGN BODY (INTRAOCULAR OR INTRAORBITAL)	**74%**	**2.6**	**InterQual**
PNEUMOTHORAX	**11%**	**4.6**	**InterQual**
SNAKEBITE	**42%**	**1.6**	**InterQual**
SUBARACHNOID HEMORRHAGE	**11%**	**4.8**	**InterQual**
SUBDURAL HEMATOMA	**11%**	**5.4**	**InterQual**

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:

1:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.
For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.

- **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.
Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.
Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.
For further clarification, see the Acute Level of Care Review Process.

3:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

4:

Mistreatment of older individuals typically presents as multiple suspicious physical exam findings rather than an isolated finding. Differentiating injury patterns in abuse of older patients from non-intentional injuries such as falls can be challenging; however, abuse-related injuries have been found to occur most often on the head, neck, and upper extremities. Signs of neglect can include, but are not limited to, malnutrition, dehydration, pressure sores, poor body or oral hygiene, and dirty clothes (American College of Surgeons Trauma Quality Programs, 2019 [cited Jun 27 2023]).
Instruction: While the literature focuses on the abuse of older individuals, these criteria can be applied to any adult-caregiver relationship.

5:

Instruction: The abuse or neglect criteria are intended to capture patients with minor injuries that are not included in the General Trauma subset; however, these patients require admission to determine if the current living arrangements are safe. A higher level of care may be needed based on the injury findings or required interventions.

6:

Instruction: This criterion is intended for patients requiring a foreign body removal procedure that cannot be performed in the emergency department (e.g., endoscopy) and cannot be done immediately for reasons such as operating room or provider availability.

7:

Monitoring after foreign body retrieval may be necessary and is at the discretion of the provider. Guidelines recommend that patients be considered for hospitalization for monitoring after endoscopic therapy if the foreign body extraction is difficult, if there is ingestion of multiple objects or foreign bodies that pose a high risk for complication (e.g., sharp-pointed objects, batteries, magnets, objects larger than 5-6 cm), or if there is extensive mucosal injury resulting from the foreign body ingestion or endoscopic treatment (Birk et al., Endoscopy 2016, 48: 489-96). Complications may include perforation, bleeding, infection, and/or airway compromise. Following endoscopic foreign body removal, complications can occur at a rate of up to 5%(0.05), with factors such as the size, surface, and type of foreign body increasing the risk (Rammohan et al., Cureus 2023, 15: epub).

8:

Monitoring may include, but is not limited to, vital signs monitoring (e.g., heart rate, blood pressure, oxygen saturation), serial physical exams, or serial laboratory studies.

9:

Although not all snakebites contain venom, it is recommended that even patients with no initial clinical evidence of envenomation undergo a 24-hour period of observation (Benjamin et al., J Spec Oper Med 2020, 20: 43-74). Repeat exams and laboratory evaluations are carried out to monitor for envenomation symptoms. Toxins released may vary widely depending on the species of snakes in a specific geographic region. Toxins may be cytotoxic, causing pain and worsening edema which can lead to neurovascular compromise; hematotoxic, causing consumption coagulopathy and bleeding; and/or neurotoxic, causing paralysis and respiratory insufficiency. Other symptoms that may be associated with snakebites include muscle necrosis, rhabdomyolysis, and acute kidney injury (Seifert et al., N Engl J Med 2022, 386: 68-78; Knudsen et al., Front Immunol 2021, 12: 661457).

10:

Venomous snakebites may cause local effects such as swelling, blistering, bruising, and necrosis at the bite site. Antivenom treatment is indicated if local symptoms progress rapidly (Ralph et al., Bmj 2022, 376: e057926).

11:

Coagulopathy in the setting of a snakebite is an indication to administer antivenom. However, if a patient is otherwise asymptomatic, it is recommended that coagulation studies (e.g., fibrinogen, prothrombin time (PT), partial thromboplastin time (PTT), and international normalized ratio (INR)) be repeated to confirm abnormalities prior to administering antivenom (Benjamin et al., J Spec Oper Med 2020, 20: 43-74).

12:

A neurological assessment establishes a baseline so that subtle changes can be monitored. A comprehensive neurological assessment often includes an evaluation of:

- Mental status (e.g., level of consciousness, orientation, insight, calculation ability)
- Cranial nerves (e.g., olfactory, optic, vestibulocochlear)
- Motor system (e.g., muscle tone, strength, reflexes)
- Sensory system (e.g., light touch, pain, temperature)
- Coordination (e.g., orchestration and fluidity of movement)
- Gait (e.g., heel-to-toe straight-line walking)

13:

A neurovascular assessment is done to identify any signs of neurovascular compromise and should include the following:

- Evaluation of the quality of the peripheral pulses and capillary refill rate
- Pain assessment
- Assessment for warmth in the extremity
- Gross motor and sensory function of fingertips and toes

Presence of any of the following signs suggests neurovascular compromise: painful passive motion, pallor of the extremity, paresthesias, paralysis, pulselessness or diminished pulses, or poikilothermia (fluctuating temperature) (Gerhard-Herman et al., Circulation 2017, 135: e686-e725).

14:

Blunt cardiac injury can result from chest wall trauma, which can lead to a wide range of symptoms, from mild asymptomatic cardiac contusion to fatal myocardial rupture. Patients can also sustain septal and/or valvular injuries and myocardial infarction. Cardiac contusion can lead to pain, electrocardiogram (ECG) abnormalities, arrhythmias, contractile abnormalities, heart failure, and ischemia. Even patients with seemingly silent and subtle ECG abnormalities may need close monitoring, as these may be signs of a more severe underlying cardiac injury (El-Qawaqzeh et al., J Surg Res 2023, 281: 22-32; Nair et al., J Cardiothorac Surg 2023, 18: 71; Williams et al., BJA Educ 2020, 20: 332-40).

15:

Instruction: Patients with chest wall trauma and suspected blunt cardiac injury can be reviewed here if their electrocardiogram (ECG) and troponin are normal or at baseline. Patients with abnormal ECG and/or troponins should be reviewed at a higher level of care.

16:

An electrocardiogram (ECG) alone is insufficient to definitively rule out blunt cardiac injury (BCI); however, when combined with troponin, the sensitivity of ruling out BCI has been reported to be 87.5%(0.875) (Kyriazidis et al., World J Emerg Surg 2023, 18: 36). The Eastern Association for the Surgery of Trauma (EAST) recommends both ECG and troponin be done to rule out BCI (Clancy et al., J Trauma Acute Care Surg 2012, 73: S301-6). However, the timing of these measurements has yet to be determined. Therefore, patients with suspected BCI who have a negative ECG and troponin on presentation may benefit from a period of observation for continuous cardiac monitoring.

17:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

18:

When minor trauma occurs in a pregnant individual with a viable pregnancy, close monitoring is recommended for at least 4 hours to allow time to rule out serious complications (e.g., placental abruption, preterm labor). The presence of uterine tenderness, abdominal pain, vaginal bleeding, sustained contractions, or abnormal fetal heart patterns may require a longer period of observation (Greco et al., Obstet Gynecol 2019, 134: 1343-57; Jain et al., J Obstet Gynaecol Can 2015, 37: 553-74).

19:

Instruction: If specific maternal injuries not related to pregnancy are identified, such as fractures or solid organ injuries, patients can be reviewed under those specific injuries in the Trauma subset if available. If pregnancy complications are identified, such as placental abruption or uterine rupture, patients can be reviewed under the Antepartum subset.

20:

The fetal heart rate (FHR) fluctuates constantly in response to changes in the intrauterine environment and other stimuli (e.g., uterine contractions). Surveillance of FHR and FHR patterns is used to assess fetal well-being and oxygenation to identify potential uteroplacental and fetal compromise, allowing for intervention, prevention, and mitigation of adverse outcomes (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27; Devane et al., Cochrane Database Syst Rev 2017, 1: CD005122). Standard FHR monitoring techniques include auscultation, Doppler ultrasound, and electronic fetal monitoring that captures the FHR on a graphical printout called a cardiotocograph using an ultrasound transducer externally attached to the maternal abdomen or an electrode placed directly on the fetal scalp internally (Devane et al., Cochrane Database Syst Rev 2017, 1: CD005122). FHR monitoring is a component of the nonstress test, contraction stress test, biophysical profile, and modified biophysical profile (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27).

21:

Tocography is used to detect uterine contractions and is most commonly performed externally, although internal monitoring may be indicated in select cases. External tocography, which records uterine activity, involves the placement of a transducer over the uterine fundus.

22:

Instruction: Patients with suspected injuries to the liver, spleen, kidneys, ureter or urethra, pancreas, or intestines can be reviewed using this criterion.

23:

Patients with abdominal trauma but no obvious signs and symptoms of organ injury or an equivocal computed tomography (CT) finding may need further observation. For example, a small amount of intra-abdominal fluid on CT without obvious solid or hollow organ injury could indicate a small or occult injury, and patients with such findings may require serial exams and/or laboratory studies to evaluate the need for repeat imaging and possible surgical intervention (Brenner and Hicks, Emerg Med Clin North Am 2018, 36: 149-60). Bowel injuries remain one of the most commonly missed injuries, even with CT, and a high index of suspicion is necessary especially if a seatbelt sign is present (Smyth et al., World J Emerg Surg 2022, 17: 13). In addition, some injuries, such as pancreatic, hollow organ, or mesenteric injuries, may take time to manifest themselves. An increase in lipase and/or amylase levels may be linked to pancreatic and hollow viscus organ injuries. If these levels continue to increase or the patient experiences persistent abdominal pain, further diagnostic tests such as repeat CT scans may be required (Smyth et al., World J Emerg Surg 2022, 17: 13; Leenellett and Rieves, Emerg Med Clin North Am 2021, 39: 795-806; Coccolini et al., World J Emerg Surg 2019, 14: 56).

24:

Fractures occurring in the first rib are often linked to high-impact trauma since they are well supported by other structures like the scapula, clavicle, and surrounding musculature. Additionally, first rib fractures may cause a vascular injury, affecting the subclavian vessels, or a nerve injury, impacting the brachial plexus (Williams et al., BJA Educ 2020, 20: 332-40; Marro et al., Emerg Radiol 2019, 26: 557-66).
Instruction: If injuries to underlying structures are identified (e.g., lung contusion, aortic or myocardial injury), the patient should be reviewed at a higher level of care under the injury sustained or intervention required.

25:

Sternal fractures often occur during high-energy traumatic events and can result in damage to underlying cardiovascular structures (Williams et al., BJA Educ 2020, 20: 332-40; Marro et al., Emerg Radiol 2019, 26: 557-66).
Instruction: If injuries to underlying structures are identified (e.g., lung contusion, aortic or myocardial injury), the patient should be reviewed at a higher level of care under the injury sustained or intervention required.

26:

Chest wall fractures may include rib(s), sternal, or scapula fractures.

27:

The presence of cardiopulmonary comorbidities in the setting of chest wall fractures has been associated with higher mortality (Williams et al., BJA Educ 2020, 20: 332-40; Van Vledder et al., Eur J Trauma Emerg Surg 2019, 45: 575-83).

28:

Spinal trauma may involve cervical, thoracic, and/or lumbar spine.

29:

Instruction: This criterion applies to patients who require surgical consultation (e.g., neurosurgeon, orthopedic surgeon, maxillofacial surgeon) to be cleared for discharge or to determine if further treatment is warranted. If the surgical consult reveals that surgical intervention is needed, refer to the General Surgical subset for the subsequent review.

30:

Instruction: The head trauma criteria at the Observation level of care is for the review of patients with mild traumatic brain injury (TBI), commonly defined as a Glasgow Coma Scale (GCS) score of 13-15 (National Institute for Health and Care Excellence (NICE), Head injury: assessment and early management. 2023). Admission to this level of care for neurological monitoring may be warranted in patients who are at risk for complications. Mild TBI can be complicated by intracranial injury (Wintermark et al., J Am Coll Radiol 2015, 12: e1-14). However, the Observation level of care is appropriate for patients with no acute intracranial findings on head computed tomography (CT) (i.e., negative head CT). Patients with acute intracranial findings on head CT should be reviewed at the Critical level of care.

31:

A head computed tomography (CT) is considered negative if there are no acute intracranial findings.

32:

Glasgow Coma Scale (GCS) is a tool commonly used to measure neurological function. GCS evaluates the level of consciousness in acutely ill patients. A patient's GCS is the combined total score accumulated through the evaluation of the following 3 areas (ranging from 3-15) (Chou et al., In: Glasgow Coma Scale for Field Triage of Trauma: A Systematic Review. 2017):

- (Eye opening - 4 = Spontaneous, 3 = To voice, 2 = To pain, 1 = None
- Verbal response - 5 = Oriented, 4 = Confused, 3 = Inappropriate, 2 = Incomprehensible sounds, 1 = None
- Motor response - 6 = Obeys verbal commands, 5 = Localizes to pain, 4 = Withdraws to pain, 3 = Flexion response to pain, 2 = Extension response to pain, 1 = None

Of note, age may impact the relationship between anatomic injury severity and GCS scores, such that the same GCS score may correspond to a greater severity of traumatic brain injury (TBI) in adults aged 65 and older than in those younger than 65 (Salottolo et al., JAMA Surg 2014, 149: 727-34).

33:

Patients who do not return to a Glasgow Coma Scale (GCS) score of 15 or their pre-injury baseline, independent of imaging results, should be closely monitored (National Institute for Health and Care Excellence (NICE), Head injury: assessment and early management. 2023).

34:

Guidelines suggest that routine admission and repeat imaging are not required for patients with minor head injuries who are taking anticoagulants or antiplatelet medication other than aspirin if the patient is at their baseline neurologic status and their initial head computed tomography (CT) is without hemorrhage. However, due to insufficient studies examining the risk of delayed intracranial hemorrhage in individuals taking aspirin, a recommendation on whether a patient should be observed longer and undergo repeat imaging could not be made for that specific antiplatelet agent (Valente et al., Ann Emerg Med 2023, 81: e63-e105). In addition, there are insufficient studies examining the risk of delayed intracranial hemorrhage in patients taking newer antiplatelet agents (e.g., prasugrel, ticagrelor). Therefore, observation and repeat imaging may be considered for patients taking antiplatelet agents. Given that there are no clear guidelines on the timing of the initial head CT after head trauma, anticoagulated patients may also be considered for a period of observation even in the setting of a negative initial head CT.

35:

It a recent retrospective study, it was found that around 10%(0.10) of patients who were diagnosed with a concussion, but did not require admission on their first visit, returned to the emergency department (ED) within a few days (an average of 3.7 days) of the initial injury (Morrison et al., Cjem 2019, 21: 784-8). The most common reason for their return was a headache (66%(0.66)). Out of all the patients who returned, 42%(0.42) underwent a repeat CT scan, but the results were negative for all of them (Morrison et al., Cjem 2019, 21: 784-8). Although the study did not provide information about the admission rate of the patients who returned, the National Institute for Health and Care Excellence (NICE) guidelines recommend admitting patients who have persistent worrying symptoms (e.g., severe headache and persistent vomiting) after a minor head injury (National Institute for Health and Care Excellence (NICE), Head injury: assessment and early management. 2023).

36:

Persistent symptoms requiring monitoring at the Observation level of care may include vomiting, drowsiness, and/or a severe headache.

37:

In some situations, performing a complete neurological exam can be challenging due to the presence of coexisting conditions. Moreover, some patients may not be able to recognize delayed symptoms of a head injury, such as a late intracranial bleed. This patient population may include those who are experiencing alcohol or drug intoxication, cognitive impairment, or a neurodegenerative motor disorder such as Parkinson's disease (National Institute for Health and Care Excellence (NICE), Head injury: assessment and early management. 2023; American College of Surgeons Trauma Quality Programs, 2015 [cited Jun 12 2023]).
Instruction: This criterion is appropriate to select if a full neurological assessment was unable to be completed because of lack of cooperation or a coexisting condition. A period of observation may be warranted to ensure there is no decline in neurological status.

38:

Difficult neurological assessment refers to the inability to complete a full neurologic assessment due to a lack of cooperation or coexisting condition.

39:

Clinically stable patients who have experienced a small/stable pneumothorax can be discharged home based on clinical judgment and repeated chest x-ray excludes progression of the pneumothorax and symptoms are resolving. Patients requiring the placement of a small bore (≤ 14F) chest tube whose symptoms have resolved may be discharged after stable repeat chest x-ray. Frequently, these patients are otherwise young and healthy with no other comorbidities or underlying lung disease. If discharged with a small bore chest tube, these patients require thorough discharge instructions, chest tube or Heimlich valve care instruction and a follow up plan of care (Sale et al., Respir Med 2020, 166: 105931; Tschopp et al., Eur Respir J 2015, 46: 321-35).

40:

Pneumothorax (PTX) can be classified as occult or overt. In an occult PTX, it is not visible on chest x-ray (CXR) but can be detected using computed tomography (CT) or ultrasound. Conversely, overt PTX can be seen on CXR. The decision to insert a chest tube for PTX depends on the overall clinical picture and the provider's decision. However, small, or occult PTX (e.g., less than or equal to 35mm on CT or less than or equal to 20%(0.20) on CXR) can be observed without invasive intervention (de Moya et al., J Trauma Acute Care Surg 2022, 92: 103-7; Bou Zein Eddine et al., J Trauma Acute Care Surg 2019, 86: 557-64).

41:

Instruction: These criteria are to be used for a traumatic pneumothorax that occurs as a result of a physical chest injury either penetrating (open) or nonpenetrating (closed) or following a surgical procedure. A spontaneous pneumothorax should be reviewed in the General Medical subset.

42:

- Small bore chest tubes are ≤ 14F
- Large bore chest tubes are > 14F

43:

Small bore chest tubes are less than or equal to 14F and large bore chest tubes are greater than 14F (Mummadi et al., Ann Emerg Med 2020, 76: 88-102; Chang et al., Chest 2018, 153: 1201-12).

44:

In a hemodynamically stable patient, a small hemothorax may be managed without an invasive procedure. Repeat imaging can be used to monitor stability (Zeiler et al., Clin Pulm Med 2020, 27: 1-12).

45:

Burns are classified by how deeply and severely they penetrate the skin's surface. The three classifications of burns are (Young et al., Phys Med Rehabil Clin N Am 2019, 30: 111-32):
Superficial (first-degree) burns damage only the top (or epidermal) layer of the skin and heal without scar formation, contractures, or pigmentation changes. The area will blanch with pressure and can be associated with inflammation, pain, swelling, and redness. Common examples of this type of burn are sunburn and a brief scalding with hot liquids. Healing time is usually within 7 days and only symptomatic treatment is required.
Partial-thickness (second-degree) burns extend into the dermis and are classified as superficial partial-thickness and deep partial-thickness burns.

- Superficial partial-thickness burns involve damage through the epidermis and the papillary layer of the dermis. This burn is characterized by blister formation with weeping and is painful to touch. These burns heal in 14-21 days by epidermal regeneration and a full return of function is expected.
- Deep partial-thickness burns involve the destruction of the epidermis with the burn extending into the deep reticular layer of the dermis. These burns are characterized as painful with the wound appearing as red with white patches and large blisters. Healing time is usually within 3 to 5 weeks in the absence of burn infection. Hypertrophic and keloid scars are a frequent consequence of deep-partial-thickness burns. Stevens-Johnson syndrome, toxic epidermal necrolysis, and severe frostbite are included here because of the similar destruction of the epidermis on presentation.

Full-thickness (third-degree) burns are defined as complete destruction of the epidermis and the entire dermis. Healing occurs at the wound margins and by grafting. A full-thickness burn lacks pain sensation due to the damaged nerve endings, which is often a sign that distinguishes a full-thickness burn from a partial-thickness burn. Additionally, a full-thickness burn that results from an incineration-type exposure and electrical burns can damage the fascia, muscle, or bone and can be life-threatening. Extensive surgery may be required to remove the necrotic tissue. In addition to skin grafts, local or regional flaps may be necessary to cover the charred area.

46:

Effective burn management requires close follow-up to address wounds, scar management, and the physical and psychological well-being of patients. It is important to regularly monitor wounds until they are well healed (Burns 2018, 44: 1617-706). If a patient is unable to adhere to wound care or follow-up, they may require additional monitoring or education. Patients with social risk factors such as lack of food, shelter, and social support may also benefit from additional screening and support (Burns 2018, 44: 1617-706).

47:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

48:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

49:

Guidelines recommend immediate reversal of vitamin K-dependent oral anticoagulants (e.g., warfarin) with both prothrombin complex concentrate (PCC) and phytomenadione (vitamin K) in a bleeding trauma patient. For patients with life-threatening bleeding who are taking an oral direct antifactor Xa agent (e.g., apixaban, rivaroxaban), reversal with andexanet alfa is recommended. If andexanet alfa is not available, or the patient is taking edoxaban, PCC administration is recommended. For patients taking dabigatran, treatment with idarucizumab is recommended if bleeding is life-threatening (Rossaint et al., Crit Care 2023, 27: 80).

50:

Frailty is a complex condition that occurs because of the aging process and makes individuals more vulnerable to stress, leading to adverse health outcomes like impaired mobility, falls, hospitalizations, and even mortality (Vahedi et al., Int J Geriatr Psychiatry 2022, 37: epub; Kwak and Thompson, Sports Med Health Sci 2021, 3: 1-10; Hoogendijk et al., Lancet 2019, 394: 1365-75). Studies have shown that older individuals identified as frail during hospital admission are at greater risk of in-hospital mortality, longer hospital stays, and functional decline at hospital discharge (Boucher et al., EClinicalMedicine 2023, 59: epub; Cunha et al., Ageing Res Rev 2019, 56: epub). Although there are many validated frailty screening tools available, there has yet to be one that is recognized as the gold standard for identifying frailty. Most of these tools are based on two models: the frailty phenotype and the deficit accumulation model. The frailty phenotype model focuses on the physical aspects of frailty (e.g., strength, walking speed, weight loss), while the deficit accumulation model identifies the number of deficiencies (e.g., functional, cognitive, comorbidities) to calculate a frailty index score (Hoogendijk et al., Lancet 2019, 394: 1365-75). However, both models have limitations in their implementation within the acute care setting, as the frailty phenotype assessment involves performing tests to evaluate the domains (e.g., grip strength, walking speed), and the deficit accumulation model is time-consuming. Therefore, frailty criteria have been developed based on literature and the two models mentioned above to help identify older adults who may have or be at risk for frailty.

51:

In comparison to injury severity and age, frailty has been reported to be a more reliable predictor of adverse outcomes and mortality in older adult trauma patients (Alqarni et al., Age Ageing 2023, 52: epub).

52:

Instruction: The frailty criteria are intended to review the high-risk, frail trauma patients with minor injuries who would not otherwise be reviewed at the Acute level of care. A higher level of care may be needed based on the injury findings or required interventions.

53:

Instruction: This criterion applies to patients who have previously been diagnosed with frailty and are currently experiencing the condition. This information can be retrieved from the medical record, such as the problem list or past medical history. However, describing a patient's appearance as "frail" in a progress note cannot be reviewed in this criterion.

54:

Instruction: The following criteria should be selected based on the patient's baseline or current health status. A patient’s baseline frailty state can affect their risk for severe illness. Similarly, if a patient presents with signs of frailty in the setting of an acute illness or injury, it should also be seen as a marker of illness severity (Theou et al., Clin Geriatr Med 2018, 34: 369-86).

55:

Studies have suggested that incorporating cognitive impairment evaluation with physical characteristics enhances the predictability of frailty (Avila-Funes et al., J Am Geriatr Soc 2009, 57: 453-61). More recently, a systematic review and meta-analysis found that mean Mini-Mental State Examination (MMSE) scores were significantly lower in individuals who were identified as frail by the phenotype model, which focuses on physical function, compared to non-frail and pre-frail individuals. As a result of the strong relationship between physical frailty and cognitive impairment, the authors concluded that cognitive function should be incorporated into frailty assessments (Vahedi et al., Int J Geriatr Psychiatry 2022, 37: epub). Multiple frailty screening tools developed using the cumulative deficit approach assess the cognitive domain, specifically focusing on cognitive impairment or dementia (Mak et al., J Gerontol A Biol Sci Med Sci 2022, 77: 2311-9; Pajewski et al., J Gerontol A Biol Sci Med Sci 2019, 74: 1771-7; Clegg et al., Age Ageing 2016, 45: 353-60).

56:

Multimorbidity refers to the presence of 2 or more chronic conditions or diseases in an individual. Chronic conditions or diseases are those that last for a prolonged period (e.g., 1 year or more) or are permanent and result in disability, decreased quality of life, or require prolonged treatment or rehabilitation (Centers for Disease Control and Prevention, About Chronic Diseases. 2022 [cited Nov 8 2023]; Nicholson et al., J Clin Epidemiol 2019, 105: 142-6; Calderón-Larrañaga et al., J Gerontol A Biol Sci Med Sci 2017, 72: 1417-23). Examples of chronic conditions that have been identified in the literature include but are not limited to (Calderón-Larrañaga et al., J Gerontol A Biol Sci Med Sci 2017, 72: 1417-23):

- Chronic lung disease (e.g., COPD, asthma)
- Chronic kidney disease (e.g., stage 3-4)
- End-stage kidney disease
- Diabetes
- Hypertension
- Congestive heart failure
- Inflammatory bowel disease (e.g., Crohn's disease, ulcerative colitis)
- Mental health disorders (e.g., drug or alcohol dependence, depression, bipolar affective disorder)

57:

According to a recent systematic review and meta-analysis, 72%(0.72) of frail individuals have multimorbidity. The study also found a significant association between frailty and having two or more diseases (Vetrano et al., J Gerontol A Biol Sci Med Sci 2019, 74: 659-66).

58:

According to a study conducted at a Medicare Accountable Care Organization (ACO), the presence of one functional deficit on a frailty index tool in the electronic health record (EHR) was found to be independently associated with a higher risk of hospitalization and injurious falls (Pajewski et al., J Gerontol A Biol Sci Med Sci 2019, 74: 1771-7). When looking specifically at the association between hearing and vision deficits and frailty, a systematic review and meta-analysis found that hearing and vision impairment were associated with 2.5- and 3.0-fold higher odds of frailty, respectively (Tan et al., J Gerontol A Biol Sci Med Sci 2020, 75: 2461-70).

59:

Functional impairment refers to an individual's ability to independently perform activities of daily living (ADLs) (e.g., bathing, dressing, toileting, ambulating), instrumental activities of daily living (IADLs) (e.g., managing medications, finances, preparing meals, usage of the telephone, performing basic housework), or a deficit in hearing or vision that is not corrected by adaptive equipment (e.g., hearing aid, glasses).

60:

Malnutrition has been found to be closely associated with frailty (Lorenzo-López et al., BMC Geriatr 2017, 17: 108). This criterion can be met by one or more of the following:

- Documented weight loss, unintentional, of ≥ 10 pounds or ≥ 5%(0.05) of body weight in the past year (Fried et al., J Gerontol A Biol Sci Med Sci 2001, 56: M146-56)
- Score of ≤ 11 on the Mini Nutritional Assessment-Short Form (MNA-SF) (O'Caoimh et al., Int J Environ Res Public Health 2019, 16: epub; Lorenzo-López et al., BMC Geriatr 2017, 17: 108)
- Body mass index (BMI) < 18.5 kg/m^2 (Mak et al., J Gerontol A Biol Sci Med Sci 2022, 77: 2311-9; Pajewski et al., J Gerontol A Biol Sci Med Sci 2019, 74: 1771-7)
- Documented moderate or severe subcutaneous fat or muscle loss (Cederholm et al., Clin Nutr 2019, 38: 1-9; White et al., JPEN J Parenter Enteral Nutr 2012, 36: 275-83)

61:

Several studies have shown that individuals with polypharmacy are more likely to be frail (Gutiérrez-Valencia et al., Br J Clin Pharmacol 2018, 84: 1432-44). In a recent systematic review and meta-analysis of 66 studies, it was found that 59%(0.59) of older adults with frailty had polypharmacy, defined as the use of five or more medications at the same time. The study also revealed that the prevalence of polypharmacy was highest among older adults with frailty in the hospital setting, with a prevalence of 71%(0.71) (Toh et al., Ageing Res Rev 2023, 83: 101811).

62:

Chronic medications refer to prescription medications that are taken according to a schedule for at least 90 days, as opposed to being taken on an as-needed basis (Masnoon et al., BMC Geriatr 2017, 17: 230). This excludes topical medications, such as creams, or over-the-counter medications.

63:

Compartment syndrome occurs when the pressure within a fascial or osteofascial compartment increases, leading to impaired microcirculatory perfusion at first. As the pressure rises, lymphatic, capillary, and tiny venule flow decreases, followed by a compromise of venous and then arterial blood flow. This can ultimately result in tissue ischemia and necrosis. Various factors can increase the risk of developing compartment syndrome, including blunt soft tissue injuries, burns, fractures, intramuscular hematomas, and snake bites. The earliest and most common symptom is severe pain, which is usually disproportionate to the examination. Patients often describe this pain as intense, deep, and worsened by passive stretching. However, pain is subjective and may not always be a reliable indicator. On examination, the patient may display signs of compartmental tenseness or hardness, edema of the affected limb, localized motor or sensory impairments, or reduced pulse or capillary refill time. Although signs and symptoms can aid in determining a diagnosis, they are not always conclusive. The diagnosis is confirmed by monitoring the pressure within the compartment. The definitive treatment involves compartment release through fasciotomy (Long et al., J Emerg Med 2019, 56: 386-97).

64:

Eye injuries that can be reviewed in this criterion may include:

- Open globe rupture
- Intraocular foreign body
- Intraorbital foreign body

65:

Intraocular foreign bodies (IOFBs) are foreign items that pierce the ocular globe wall and become lodged in the eye. IOFB injuries are classified as open-globe injuries (OGIs) and can cause varying degrees of damage to ocular structures (Liang et al., J Ophthalmol 2021, 2021: 9933403). The damage can occur immediately or over time due to other complications such as retinal detachment or endophthalmitis. Research has shown that a primary repair of OGIs within 12 to 24 hours of the injury onset results in better outcomes and lowers the risk of endophthalmitis.

The goal of primary repair is globe closure, and the IOFB may or may not be removed at that time. When there is a risk of endophthalmitis, immediate removal is recommended, especially for organic materials such as wood or plants. However, inert materials such as glass or plastic may be left in place until secondary repair (Zhou et al., Clin Ophthalmol 2022, 16: 2545-59).

66:

Intraorbital foreign bodies occur infrequently and can either be asymptomatic or cause serious problems such as cellulitis, optic neuropathy, or ocular dysmotility. The decision to remove an intraorbital foreign body depends on the material, location, presentation, and level of impairment. Inorganic foreign bodies that are asymptomatic can be monitored without surgery. However, it is recommended to remove organic foreign bodies (e.g., wood, plastic) due to the risk of infection (Chai et al., Int J Ophthalmol 2023, 16: 1130-7).

67:

Instruction: Patients with severe injuries requiring more invasive interventions than listed here should be reviewed at a higher level of care. In polytrauma patients, criteria should be reviewed at the level corresponding to the most severe injury.

68:

Some penetrating injuries (e.g., stab or gunshot wounds) may be managed nonoperatively. If a patient has an anterior penetrating wound and the local wound exploration (LWE) shows no signs of injury, they may be discharged from the emergency department without the need for a computed tomography (CT) scan. Those with equivocal or positive LWE, which could mean that the peritoneum has been penetrated, may need close monitoring for changes in their abdominal exam and/or laboratory evaluation. If there is any significant drop in hemoglobin, additional studies or interventions, such as CT scans or emergent surgical exploration, may be required (Smyth et al., World J Emerg Surg 2022, 17: 13; Brenner and Hicks, Emerg Med Clin North Am 2018, 36: 149-60).

69:

Organ injury may include injury to the liver, spleen, kidneys, ureter or urethra, pancreas, or intestines.

70:

Regional anesthesia may be used to control pain in patients with chest wall fractures, especially rib fractures. Intercostal nerve blocks, paravertebral blocks, and thoracic epidural catheters are some of the frequently used regional anesthesia techniques (Williams et al., BJA Educ 2020, 20: 332-40; Dennis et al., Surg Clin North Am 2017, 97: 1047-64). Regional anesthesia aids in pain management, improving respiratory function and inspiratory lung volumes. Proper pain control, along with therapies to aid lung expansion and mobility are important in managing chest wall injuries to prevent pulmonary complications such as pneumonia (Dennis et al., Surg Clin North Am 2017, 97: 1047-64).

71:

Respiratory compromise in patients with rib fractures may manifest as hypoxia or an inability to perform incentive spirometry greater than 1000 mL (or greater than 15 ml/kg) (Brasel et al., J Trauma Acute Care Surg 2017, 82: 200-3; Dennis et al., Surg Clin North Am 2017, 97: 1047-64).

72:

Patients over the age of 65 are considered high risk for rib fracture complications (Coary et al., Age Ageing 2020, 49: 161-7; Van Vledder et al., Eur J Trauma Emerg Surg 2019, 45: 575-83). Older individuals who suffer from chest wall injuries often experience more severe outcomes, including longer time on a ventilator, higher rates of pneumonia, and acute respiratory distress syndrome (ARDS) compared to patients who are younger (Dennis et al., Surg Clin North Am 2017, 97: 1047-64). According to the Western Trauma Association, patients over the age of 65 with more than 2 rib fractures should be closely monitored in an intensive care setting. However, if the injury is relatively mild (e.g., minimally displaced) and there are no other comorbidities, a lower level of care may be appropriate (Brasel et al., J Trauma Acute Care Surg 2017, 82: 200-3).

73:

Dyspnea in patients with rib fractures has been associated with a higher likelihood of developing respiratory complications and prolonged hospital stays (Leichtle et al., Am Surg 2020, 86: 1194-9).

74:

Upper facial fractures can lead to cerebral spinal fluid (CSF) leaks due to the proximity of the frontal bones with the neighboring cerebral structures. Patients with minor CSF leaks may initially be managed with conservative treatments such as head elevation, stool softeners, blood pressure control, and bed rest to allow the dural defect to heal spontaneously. Most small CSF leaks resolve on their own, but if they persist, surgical intervention may be considered (Chen et al., Facial Plast Surg 2023, 39: 253-65).
Instruction: Those requiring a lumbar drain should be reviewed at the Critical level of care in this subset, while those requiring surgical repair should be reviewed in the General Surgical subset.

75:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

76:

Maxillofacial trauma can cause airway obstruction due to various factors, including swelling and edema of the soft tissues. Although the airway compromise is usually evident at the time of injury, it may also present later, as edema can worsen within 48 to 72 hours post-injury (Esonu and Sardesai, Semin Plast Surg 2021, 35: 225-8; Barak et al., Biomed Res Int 2015, 2015: 724032).

77:

Instruction: Patients with compression fractures or sacral fractures without pelvic ring disruption can be reviewed in this criterion only if the fracture occurred as a result of a traumatic injury (e.g., fall).

78:

It is recommended that regional nerve blocks be used for acute hip fracture pain upon presentation. Additionally, nerve blocks should be considered for post-operative pain management in all fractures that require surgery. If a nerve block is chosen as the preferred method of pain control, a continuous catheter should be considered over a single-shot block to ensure optimal pain relief and to reduce the likelihood of rebound pain (Hsu et al., J Orthop Trauma 2019, 33: e158-e82).

79:

Long bone fractures have been reported to be associated with acute blood loss, which may result in the need for a blood transfusion. A recent cohort study, including 293 patients with isolated femoral fractures, found that 36% (0.36) required a blood transfusion, with 15%(0.15) receiving the transfusion pre-operatively (Wertheimer et al., Injury 2018, 49: 846-51). The authors also found that admission hemoglobin level was independently linked to the need for blood transfusion within the first 48 hours of admission.

80:

A halo device used for fixation or traction, as well as Crutchfield cervical tongs, pelvic, or skeletal traction that is continuous or cyclic can be reviewed in this criterion.

81:

Basilar skull fractures involve at least one of the five bones that make up the base of the skull and are associated with several complications; therefore, neurologic monitoring should be considered for patients who sustain this injury. Basal fractures frequently cross foramen, causing injury to cranial nerves and blood vessels. In addition, because the dura mater is tightly attached to the skull base, the shattered bone is prone to lacerating it, resulting in a cerebrospinal fluid (CSF) leak (Naidu et al., Interdisciplinary Neurosurgery: Advanced Techniques and Case Management 2020, 23: epub). Basilar skull fractures also put patients at risk for intracranial hematoma. A retrospective study conducted on autopsy cases to determine if the head injury impact site correlated with subdural and epidural hematomas found that when the place of impact was the temporal region, which has close proximity to the middle meningeal artery and vein, there was a greater likelihood of subdural and epidural hematomas

(Evaggelakos et al., J Forensic Leg Med 2022, 85: 102283).

82:
The Glasgow Coma Scale (GCS) can be used to define the severity of a traumatic brain injury (TBI). Mild TBI corresponds to a GCS of 13 to 15, moderate TBI corresponds to a GCS of 9 to 12, and severe TBI corresponds to a GCS of 8 or less (National Institute for Health and Care Excellence (NICE), Head injury: assessment and early management. 2023; American College of Surgeons Trauma Quality Programs, 2015 [cited Jun 12 2023]).

83:
Instruction: These criteria are meant to include patients who have had a chest tube placed during their current admission, not for patients who had a chest tube placed during a prior episode of care.

84:
The severity of a drowning injury is classified into 6 grades (Szpilman and Morgan, Chest 2021, 159: 1473-83):
- Grade 1: Responsive, positive cough with normal lung auscultation
- Grade 2: Responsive, rales in some pulmonary fields, and may require low-flow oxygen
- Grade 3: Responsive, rales auscultated in all pulmonary fields (acute pulmonary edema), normotensive, and requiring high flow oxygen or mechanical ventilation
- Grade 4: Responsive, rales auscultated in all pulmonary fields (acute pulmonary edema), hypotensive, and requiring high flow oxygen or mechanical ventilation
- Grade 5: Unresponsive, pulse present, requires artificial ventilation
- Grade 6: Unresponsive, pulse absent, no obvious physical signs of death, cardiopulmonary resuscitation

Instruction: Patients requiring advanced respiratory support (e.g., high-flow nasal cannula, non-invasive positive pressure ventilation, mechanical ventilation) should be reviewed at a higher level of care.

85:
Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO_2) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO_2 with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO_2 retention, where a precise inspired FiO_2 is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO_2. The following are estimates of O_2 delivered at the associated flow rates:
LOW-FLOW SYSTEMS
Nasal cannula
Room air 21%(*0.21*)
1L 24%(*0.24*) **4L** 36%(*0.36*)
2L 28%(*0.28*) **5L** 40%(*0.40*)
3L 32%(*0.32*) **6L** 44%(*0.44*)
Simple oxygen masks
- 35-50%(*0.35-0.50*) FiO_2 (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO_2 in the mask

Venturi masks
- 24-50%(*0.24-0.50*) FiO_2 (4-12 L/min)

Partial rebreathing masks
- 40-70%(*0.40-0.70*) FiO_2 (6-10 L/min)

Non-rebreathing masks
- 60-80%(*0.60-0.80*) FiO_2 (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS
Air-entrainment masks/nebulizers
- 24-40%(0.24-0.40) FiO_2

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO_2

86:

Pulmonary contusions are usually diagnosed by computed tomography (CT) scan, as findings may be delayed on chest x-ray. Contused lungs may lead to impaired gas exchange and can result in respiratory distress. Severe pulmonary contusions can lead to respiratory failure and acute respiratory distress syndrome (ARDS) (Dennis et al., Surg Clin North Am 2017, 97: 1047-64; Majercik and Pieracci, Thorac Surg Clin 2017, 27: 113-21).

87:

The American Burn Association guidelines for burn patient referral recommend that patients be transferred to a burn center for the following reasons (American Burn Association, 2022 [cited Jun 26, 2023]):

- Partial-thickness burns greater than or equal to 10%(*0.10*) total body surface area
- Full-thickness burns in any age group
- Partial-thickness and full-thickness burns to localized areas (e.g., face, hands, feet, genitalia, perineum, over joints)
- Inhalation injury
- Patients with concomitant trauma with an increased risk of morbidity or mortality
- Significant underlying illness complicating burns
- High voltage electrical injury, including lightening injury
- Chemical burns
- Children with burns seen in hospitals without qualified personnel or equipment for their care
- Burn injury in patients who will require special social, emotional, or long-term rehabilitative interventions
- Poorly controlled pain

88:

Instruction: Severe frostbite is included in the partial thickness criterion because it presents with similar destruction of the epidermis.

89:

Individuals over the age of 50 who suffer from even minor burns, defined as less than 10%(0.10) of their total body surface area (TBSA), are at higher risk for systemic complications. It is recommended that patients in this category should have a burn center consultation to develop a treatment plan. If the burns are superficial partial thickness or less than 5%(0.05) TBSA deep partial thickness wounds, and if the wound care can be managed at home without the need for parenteral opioids, then discharge can be considered (Vercruysse et al., J Trauma Acute Care Surg 2019, 87: 1239-43).

90:

Chemical burns are more severe when the exposure duration is prolonged and the chemical is highly concentrated. It's important to note that chemical exposure can affect more than just the skin. Some chemicals can be absorbed through the skin, such as phenol, hydrofluoric acid, and hydrogen sulfide, and cause systemic symptoms such as central nervous system toxicity or electrolyte imbalances. Additionally, concentrated inorganic acids, phenol, chlorine, and anhydrous ammonia can cause lung damage from inhalation (Burns 2016, 42: 953-1021).
Instruction: Patients requiring more intensive interventions than listed at the Acute level of care should be reviewed at a higher level of care.

91:

Inhalation injuries can harm both the upper and lower airways. The upper airway can be damaged by direct heat, while the lower airway can be affected by the chemicals present in smoke. Clinical signs that suggest an inhalation injury include singed nasal hair, soot in the nose, mouth, or sputum, and facial burns (Deutsch et al., Burns 2018, 44: 1040-51). Upper airway injuries are more concerning because of the possibility of airway obstruction due to heat damage and subsequent swelling. Early intubation is recommended to secure the airway before it is compromised. Concerning symptoms of airway obstruction include voice changes, hoarseness, and stridor. Inhalation injuries that

affect the lower airway can lead to inflammation and swelling of the airway, excess mucus production, and ventilation/perfusion mismatch. Even if the patient appears asymptomatic, hospitalization should be considered if there is a suspicion of lower airway inhalation injury due to the varying onset of respiratory symptoms. Initial chest radiography is usually normal, and bronchoscopy may be indicated to visualize the lower airways (American Burn Association, 2018 [cited Aug 25, 2023]).
Instruction: Patients requiring more intensive interventions than listed at the Acute level of care should be reviewed at a higher level of care. In addition, patients with an inhalation injury caused by exposure to carbon monoxide should be reviewed using the Carbon Monoxide Poisoning subset.

92:

The most effective treatment for deep partial- and full-thickness burns is early excision and grafting. This approach aims to prevent infection and reduce the risk of burn wound sepsis. While topical anti-infectives are recommended for most burn wounds, they may impede wound healing. Therefore, it is important to consider the benefits of infection prevention against the risk of impaired wound healing when selecting a topical anti-infective agent. Systemic anti-infective medications are not recommended since they are ineffective in preventing colonization and infection of the burn wound. Topical anti-infectives, in contrast, are applied directly to the burn wound and penetrate the eschar to prevent the development of infection (Burns 2018, 44: 1617-706). Topical anti-infectives come in many different forms such as creams, ointments, liquids, and impregnated dressings (Jeschke et al., Nat Rev Dis Primers 2020, 6: 11). The recommended agents for burn care include (Burns 2018, 44: 1617-706):

- Silver-based topical agents: appropriate for deep burns and superficial wounds that are expected to heal spontaneously
- Mafenide acetate: ideal for deep or infected burns and deep burns of the ear due to its ability to penetrate eschar and tissue. However, it is less suitable for superficial burn wounds because it is cytotoxic to keratinocytes and fibroblasts, which may impede healing
- Dankin solution and acetic acid: effective against biofilms, making them useful for chronic, heavily colonized, and infected wounds
- Topical anti-infective ointments (e.g., bacitracin, polymyxin B sulfate, neomycin): may be suitable for small superficial burns. However, evidence is limited regarding their effect on preventing wound infection and healing

93:

Complex wound treatment is typically a multi-step process. The steps involved in caring for burn wounds may include, but are not limited to (Burns 2016, 42: 953-1021):

- Premedicating with opioids and anxiolytics prior to wound care
- Staging of wound(s)
- Cleansing the wounds with gentle washing or pressurized irrigation to wash away bacterial contaminants and debris
- Mechanical debridement to remove biofilm
- Applying topical anti-infective agents
- Applying dressings (primary dressings make direct contact with the wound surface, and secondary dressings provide a layer to absorb drainage)

94:

Burn injuries cause a hypermetabolic state, putting patients at risk for malnutrition, which can lead to a weakened immune response and impaired healing. As a result, nutrition is an important component of burn treatment. The International Society for Burn Injury (ISBI) recommends that oral or enteral nutrition be initiated as soon as possible (Burns 2016, 42: 953-1021). The British Burn Association recommends early enteral feeding within 6-12 hours of injury for patients with major burns (greater than or equal to 15%(0.15) total body surface area) and for those who are unable to take an oral diet (British Burn Association, 2018 [cited Aug 31, 2023]). The enteral route is preferred over parenteral nutritional because it helps preserve gastrointestinal function and reduces bacterial translocation. However, if enteral nutrition is not tolerated or contraindicated, total parenteral nutrition may be initiated (McGovern et al., BJA Educ 2022, 22: 138-45; Clark et al., Burns Trauma 2017, 5: 11; Burns 2016, 42: 953-1021).

95:

Opioids continue to be the most common therapy for burn injury pain. The opioid of choice and the route of administration rely on regional preferences and patient characteristics, such as renal function (McGovern et al., BJA Educ 2022, 22: 138-45). The American Burn Association recommends utilizing opioids in conjunction with non-opioid (e.g., acetaminophen) and nonpharmacological measures (e.g., cognitive-behavioral therapy, hypnosis, virtual reality) rather than in isolation (Romanowski et al., J Burn Care Res 2020, 41: 1129-51).

96:

The American Burn Association recommends considering the use of ketamine for procedural sedation in adult burn patients, according to their guidelines for managing acute pain. While there is a lack of large-scale studies on the use of ketamine, several single-center studies have shown positive outcomes, such as reduced pain when combined with midazolam during burn dressing changes (Romanowski et al., J Burn Care Res 2020, 41: 1129-51).

97:

Neurovascular assessments are recommended in any extremity with a circumferential cutaneous burn or an electrical contact site, given the risk for compartment syndrome. Electrical burn injuries may result in severe injury to deep tissues, despite superficial tissues appearing normal or uninjured (American Burn Association, 2018 [cited Aug 25, 2023]).

98:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

99:

Accidental hypothermia is defined as an involuntary drop to a core temperature of 95°F(35°C) or less. It may further be staged as mild when core temperature is greater than 89.6°F(32°C) to 95°F(35°C), moderate when core temperature is between 82.4°F(28°C) to 89.6°F(32°C), and severe when core temperature is less than 82.4°F(28°C). The risk of cardiac arrest increases below 89.6°F(32°C), especially in older adults. Accidental hypothermia may occur due to prolonged exposure to cold environments, which is also known as primary accidental hypothermia. It may also occur due to other causes such as hypovolemia in major trauma patients. Hypothermia can lead to several complications including hemodynamic instability, dysrhythmias, coagulopathy, and acidosis (Paal et al., Int J Environ Res Public Health 2022, 19: epub; van Veelen and Brodmann Maeder, Int J Environ Res Public Health 2021, 18: epub).

100:

Instruction: For complications arising from hypothermia that are not addressed in the Trauma subset, refer to the appropriate condition-specific or general subset.

101:

The Eastern Association for the Surgery of Trauma (EAST) recommends that patients who may have suffered blunt cardiac injury (BCI) undergo an electrocardiogram (ECG) and have their troponin levels measured. If the test results show a new ECG abnormality (e.g., arrhythmia, ST changes, ischemia, or heart block) or elevated troponin, EAST recommends that the patient be admitted to a monitored setting for further evaluation and monitoring of troponin and cardiac rhythm (Clancy et al., J Trauma Acute Care Surg 2012, 73: S301-6).

102:

Most patients with decompression sickness respond to a single hyperbaric oxygen treatment; however, if symptoms persist, repeat daily treatments may be needed until there is a complete resolution of symptoms or until there is a lack of improvement after 2 consecutive treatments. Of those requiring further treatments, the majority will stabilize after an additional 1-2 treatments (Moon and Mitchell, Undersea Hyperb Med 2019, 46: 685-93).

103:

Hyperbaric oxygen treatment (HBOT) is recommended for all patients with symptomatic decompression sickness

(DCS) when feasible. However, patients with mild DCS experiencing symptoms such as limb pain, constitutional symptoms, subjective sensory symptoms, and rash, and without neurological symptoms, can be treated with normobaric oxygen if HBOT is not available. However, if the person does not respond to normobaric oxygen or has severe DCS, HBOT is the recommended course of action (Moon and Mitchell, Undersea Hyperb Med 2019, 46: 685-93).

104:

The high-risk trauma criteria were developed based on the national guideline for field triage of injured trauma patients. Patients involved in such traumatic events are recommended to be transferred to a trauma center as there is a moderate risk for serious injury. Anatomic and physiologic findings do not always identify patients with significant injuries, so mechanism criteria are important to identify those with high potential for injuries (Newgard et al., J Trauma Acute Care Surg 2022, 93: e49-e60). Such injuries may not be immediately apparent upon presentation, and patients who have experienced one of these traumatic events may be appropriate for monitoring at the Intermediate level of care.

105:

Vehicle telemetry provides such data as seat belt use, direction of impact, and change in velocity. Since many newer vehicles provide this valuable information, it can be extremely helpful in assisting the triage of motor vehicle trauma patients (Sasser et al., MMWR Recomm Rep 2012, 61: 1-20).

106:

Transport vehicles may include, but are not limited to, the following:

- Motorcycle
- All-terrain vehicle (ATV)
- Horse
- Electric scooter

107:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO_2. The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

108:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

109:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some

cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).
The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.
Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

110:

A retroperitoneal hematoma may have many sources including (Paun et al., Chirurgia (Bucur) 2021, 116: 725-36):

- Injury to retroperitoneal organs (e.g., kidneys, adrenals, pancreas, duodenum, ascending or descending colon, inferior rectum)
- Injury to the retroperitoneal great vessels (e.g., portal vein, renal vessels, iliac vessels)
- Injury to the small retroperitoneal vessels (e.g., mesenteric vessels)

Instruction: This criterion is intended for patients who are hemodynamically stable with evidence of retroperitoneal hematoma or vascular contusion on imaging. Patients who are hemodynamically unstable and/or require intervention (e.g., angioembolization) should be reviewed at the Critical level of care.

111:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

112:

Indications for angioembolization (AE) associated with abdominal trauma may include (Cimbanassi et al., J Trauma Acute Care Surg 2018, 84: 517-31):

- Pseudoaneurysm or any vascular injury identified on computed tomography (CT) regardless of bleeding
- Arterial blush or extravasation seen on CT
- Splenic injury grade IV or V, regardless of the absence of bleeding on CT
- Persistent hematuria in kidney injury (Coccolini et al., World J Emerg Surg 2019, 14: 54)
- Arteriovenous (AV) fistula

113:

Findings of hemodynamic instability include a sustained heart rate greater than 120/min, systolic blood pressure less than 90 mmHg, systolic blood pressure less than 110 mmHg with chronic hypertension, or decrease in systolic blood pressure greater than or equal to 30 mmHg from baseline.

114:

Aortic injury resulting in a small intimal tear or intramural hematoma may be managed nonoperatively. This approach mainly involves controlling the patient's blood pressure and heart rate with intravenous infusions of beta-blockers, calcium channel blockers, and/or vasodilators (Kapoor et al., Radiographics 2020, 40: 1834-47). For more severe injuries, such as a pseudoaneurysm, thoracic endovascular aneurysm repair (TEVAR) is recommended over open surgical repair since it carries a lower risk of complications (Soong et al., J Vasc Surg 2019, 70: 941-53.e13; Fox et al., J Trauma Nurs 2015, 22: 99-110).

115:

The severity of an electrical injury depends on various factors, such as voltage, current type, and the duration of contact. It is possible to have deeper tissue or organ injury without significant skin damage (Burns 2018, 44: 1617-706).

116:

Burn patients may experience pain, agitation, and anxiety. In severe cases, continuous drips such as ketamine or dexmedetomidine may be helpful in managing these symptoms while also reducing the need for opioids or benzodiazepines (Romanowski et al., J Burn Care Res 2020, 41: 1129-51).

117:

Extensive burns can cause increased capillary permeability, leading to edema and intravascular hypovolemia. To prevent or treat shock and end-organ damage, fluid resuscitation is crucial in these patients. However, over-resuscitation can result in more swelling and an increased risk of compartment syndrome and respiratory failure. Different formulas exist to guide the volume and rate of fluid resuscitation in severe burn patients. These formulas recommend IV fluid administration of between 2 and 4 mL/kg/burn total body surface area within the first 24 hours of injury. However, if the burn is unable to be excised or closed or if the total body surface area of the burn is large, fluid resuscitation may be necessary beyond the first 24 hours. Monitoring urine output closely is recommended to guide the titration of IV fluids to achieve a goal of 0.3 to 0.5 mL/kg/h of urine output (Jeschke et al., Nat Rev Dis Primers 2020, 6: 11; Burns 2016, 42: 953-1021).

118:

Procedural sedation is the administration of a sedative, with or without analgesia, to induce an altered state of consciousness while preserving cardiorespiratory function, to allow a patient to tolerate a painful procedure. The target level of sedation should be specific to the needs of the patient, what the procedure involves, and the duration of the procedure (Godwin et al., Ann Emerg Med 2014, 63: 247-58 e18; National Institute for Clinical Excellence (NICE), Clinical guideline 112. NICE 2010, 37p.).

119:

Escharotomy may be necessary when a circumferential or near-circumferential eschar impedes tissue circulation or circulation beyond the eschar. It may also be necessary when eschar on the neck or trunk hinders respiration (Burns 2016, 42: 953-1021).

120:

Mortality post cardiac arrest is high with most deaths occurring within the first 24 hours. Systematic post-cardiac arrest care after the return of spontaneous circulation can improve the likelihood of survival with good quality of life. Multiple organ systems are affected after cardiac arrest, thus development of a system wide proactive plan of care may be beneficial for cardiac arrest survivors.
In adult patients, post resuscitation care should include (Callaway et al., circulation 2015, 132: S465-S82):

- Therapeutic hypothermia for those patients unable to follow commands after resuscitation
- Optimization of hemodynamics and gas exchange
- Immediate coronary reperfusion with percutaneous coronary intervention (PCI) when indicated
- Glycemic control
- Neurological management and prognostication

In pediatric patients, post resuscitation care should include (Chameides et al., Pediatric Advanced Life Support: Provider Manual Pap/Crds P ed. 2012):

- Maintaining cardiovascular function and promoting adequate tissue perfusion
- Optimizing oxygenation and ventilation
- Glycemic control
- Correction of electrolyte or acid-base abnormalities
- Maintaining appropriate levels of analgesia and sedation
- Consideration of therapeutic hypothermia if coma persists following resuscitation

121:

Intracranial pressure (ICP) monitoring may be indicated in patients with a Glasgow Coma Scale (GCS) score of 8 or less and structural brain damage on computed tomography (CT) scan, as well as those with structural brain damage at risk of progression to or confirmed intracranial hypertension (e.g., large or multiple contusions), or those who have significant changes in their clinical exam or imaging (Picetti et al., World J Emerg Surg 2019, 14: 53; American College of Surgeons Trauma Quality Programs, 2015 [cited Jun 12 2023]).

122:

Hyperosmolar therapy creates an osmotic gradient, facilitating the removal of water across the blood brain barrier, resulting in a decreased brain volume and a decrease in intracranial pressure. While hypertonic saline and mannitol are the 2 most used hyperosmolar agents, there is insufficient evidence to recommend one over the other for improving neurological outcomes in patients with acute ischemic stroke (Cook et al., Neurocrit Care 2020, 32: 647-66). Doses and strengths are generally determined by institutional practice patterns. Common side effects associated with osmotic therapy include dehydration, hypotension, renal failure, and acute respiratory distress.

123:

Disseminated intravascular coagulation or DIC is a hematological emergency where there is abrupt systemic cascade activation of coagulation leading to thrombosis of midsize to small blood vessels rapidly leading to acute organ dysfunction. Simultaneously, there is consumption of platelets and proteins involved in the coagulation cascade, resulting in acute thrombocytopenia and low clotting factors. DIC can be caused by conditions such as sepsis, trauma, severe obstetrical complications, and malignancies. Laboratory abnormalities found in DIC include low fibrinogen (not always present), elevated fibrin degradation products, elevated D-dimer, thrombocytopenia, and prolonged prothrombin (PT) and activated partial thromboplastin times. Prompt therapy for DIC includes treatment against the underlying disease but may also include platelet transfusion, plasma (fresh frozen plasma; FFP) or factor transfusion in cases of hemorrhage, or anticoagulants in situations with thrombosis formation predominates (Wada et al., J Thromb Haemost 2013:).

124:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

125:

Patients who sustain high-voltage electrical injury, defined as 1000 volts or more, or exhibit abnormal electrocardiogram and/or troponin, arrhythmias, or loss of consciousness at the scene, should be closely monitored for at least 24 hours in an intensive care setting (Waldmann et al., Eur Heart J 2018, 39: 1459-65).

126:

The most frequent cardiac consequence of electric injury is arrhythmia, with sinus tachycardia and premature ventricular contractions being the most common. However, atrial fibrillation, ventricular tachycardia, and ventricular fibrillation have also been reported. Although high-voltage (e.g., lighting) or direct current exposure is likely to cause ventricular asystole, even low-voltage electricity can result in potential cardiac arrest due to ventricular fibrillation (Waldmann et al., Eur Heart J 2018, 39: 1459-65).

127:

Exposure to an electrical current can damage the muscles of the heart. Since chest pain is frequently absent, the injury may only show up as non-specific changes in the electrocardiogram (ECG), such as transient QT prolongation or ST segment and T wave abnormalities. When there is coronary artery damage or spasm, classic clinical symptoms, or ECG patterns of myocardial infarction, such as ST segment elevation or the appearance of Q waves, may be present (Waldmann et al., Eur Heart J 2018, 39: 1459-65).

128:

Instruction: Patients with suspected or confirmed acute coronary syndrome (ACS) should be reviewed in the Acute Coronary Syndrome (ACS) subset.

129:

High voltage is defined as greater than or equal to 1000 volts.

130:

An emergent cricothyroidotomy or tracheotomy can be a lifesaving procedure in patients with trauma who are unable to be intubated or ventilated due to their injuries (e.g., penetrating or blunt neck trauma, facial injuries) (Louro and Varon, Int Anesthesiol Clin 2021, 59: 10-6).

131:

Patients with inhalation injuries may require therapeutic serial bronchoscopies to remove epithelial debris and prevent airway obstruction from cast formation (Foncerrada et al., Ann Plast Surg 2018, 80: S98-s105).

132:

The mainstay of treatment of patients with hemorrhagic shock is blood product transfusion (e.g., packed red blood cells). Despite the general principle of minimizing the use of intravenous fluids in patients with known hemorrhage, crystalloids are often administered until blood products are available, especially in patients who are hypotensive (e.g., mean arterial pressure less than 65). Vital sign changes may underestimate blood loss in patients. Patients with a 30%(0.30) loss of total blood volume may have little to no change in systolic blood pressure (SBP) and only demonstrate modest tachycardia (i.e., heart rate greater than 100). By the time an SBP has dropped to less than 100 mmHg, there may be as much as a 40%(0.40) loss of total blood volume.

133:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).
Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

134:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous pressure, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

135:

An intra-aortic balloon pump (IABP) is a mechanical circulatory support device that utilizes a catheter-mounted intravascular device positioned in the descending thoracic aorta. The balloon automatically inflates during early diastole, increasing coronary blood flow and deflates during early systole, reducing afterload as the left ventricle ejects. Indications for use include cardiogenic shock, temporary support of left ventricular function after myocardial infarction (MI), preservation of myocardial viability, or refractory ventricular arrhythmias post-MI or preoperative bridge before coronary artery bypass graft (CABG) surgery. Contraindications include aortic regurgitation and aortic dissection.

136:

Instruction: This criterion includes administration of aerosolized medication via nebulizer or inhaler. Appropriate selection of a delivery device for aerosol therapy depends on multiple factors including cognitive state or developmental stage, manual dexterity, breathing pattern, respiratory drive, and drug formulation. Efficacy is optimized when the device is used correctly and complements the patient's age, physical characteristics, and cognitive ability (Tashkin, Int J Chron Obstruct Pulmon Dis 2016, 11: 2585-96; Brittan et al., J Pediatr 2015, 166: 998-1005 e1).

137:

This criterion applies to patients with a brain hematoma, contusion, or hemorrhage in one or more of the following areas:

- Epidural
- Subdural
- Intraparenchymal
- Intraventricular
- Subarachnoid

138:

Depressed skull fractures do not always require surgical intervention, but consensus indicates that surgery may be needed in case of concurrent intracranial complications (e.g., mass effect or intracranial hematoma), focal neurologic deficits, dural laceration, frontal sinus involvement, or cosmetic deformity (Stein, World Neurosurg 2019, 121: 186-92). The surgery is typically performed soon after the injury, but emergency surgery is rarely required if the depressed skull fracture is closed. However, if it is an open depressed skull fracture, immediate surgery is necessary to prevent the risk of infection, seizures, and mortality (Stein, World Neurosurg 2019, 121: 186-92).

139:

Paresis refers to a condition of muscle weakness with a reduction in voluntary movement while paralysis refers to a complete loss of movement. These criteria include, but are not limited to, paresis or paralysis of the face as well as difficulty swallowing (i.e., dysphagia).
Instruction: Patients experiencing general weakness or inability to move limbs from a non-neurologic cause are excluded from these criteria.

140:

Visual field loss includes hemianopsia, quadrantanopia, or total blindness.

141:

Critical care monitoring should be considered for all spinal cord injuries (SCI) to quickly identify systemic or neurological complications that could potentially be avoided or treated early. In addition, acute SCI is associated with potentially fatal cardiovascular consequences that necessitate increased vigilance and preventive care in the intensive care unit (ICU) (Sacino and Rosenblatt, J Neuroanaesth Crit Care 2019, 6: 222-35).

142:

For spleen, liver, and pancreas injuries that are classified as grade III or higher, and kidney injuries that are classified as grade IV or V, it is recommended to have continuous and intensive monitoring. Additionally, an operating room should be readily available in case nonoperative management (NOM) fails (Coccolini et al., World J Emerg Surg 2020, 15: 24; Coccolini et al., World J Emerg Surg 2019, 14: 56; Coccolini et al., World J Emerg Surg 2019, 14: 54; Cimbanassi et al., J Trauma Acute Care Surg 2018, 84: 517-31; Coccolini et al., World J Emerg Surg 2017, 12: 40). In cases of grade III pancreatic injury, surgical intervention is usually required. However, according to the World Society of Emergency Surgery (WSES), some patients may undergo NOM if the injury is located very proximally to the pancreatic body (Coccolini et al., World J Emerg Surg 2019, 14: 56).

143:

The American Association for the Surgery of Trauma (AAST) organ injury severity score for spleen, liver, and pancreas injuries are as follows:
Spleen (Coccolini et al., World J Emerg Surg 2017, 12: 40):
Grade I

- Hematoma: Subcapsular, < 10%(*0.10*) surface area
- Laceration: Capsular tear, < 1 centimeter parenchymal depth

Grade II

- Hematoma: Subcapsular, 10-50%(*0.10-0.50*) surface area, intraparenchymal < 5 centimeters in diameter
- Laceration: 1-3 centimeters parenchymal depth that does not involve a parenchymal vessel

Grade III

- Hematoma: Subcapsular, > 50%(*0.50*) surface area or expanding; ruptured subcapsular or parenchymal hematoma; intraparenchymal hematoma > 5 centimeters
- Laceration: > 3 centimeters parenchymal depth or involving trabecular vessels

Grade IV

- Laceration involving segmental or hilar vessels producing major devascularization (> 25%(0.*25*) of spleen)

Grade V

- Hematoma: Completely shattered spleen
- Laceration: Hilar vascular injury devascularizes spleen

Liver (Coccolini et al., World J Emerg Surg 2020, 15: 24):
Grade I

- Hematoma: Subcapsular, < 10%(*0.10*) surface area
- Laceration: Capsular tear, < 1 centimeter parenchymal depth

Grade II

- Hematoma: Subcapsular, 10-50%(*0.10-0.50*) surface area, intraparenchymal, < 10 centimeters in diameter
- Laceration: 1-3 centimeters parenchymal depth, < 10 centimeters in length

Grade III

- Hematoma: Subcapsular, > 50%(*0.50*) of surface area or expanding, ruptured subcapsular or parenchymal hematoma; intraparenchymal hematoma > 10 centimeters
- Laceration: > 3 centimeters parenchymal depth

Grade IV

- Laceration: Parenchymal disruption involving 25-75%(*0.25-0.75*) of hepatic lobe

Grade V

- Laceration: Parenchymal disruption involving > 75%(*0.75*) of hepatic lobe
- Vascular: Juxtahepatic venous injuries (e.g., retrohepatic vena cava / central major hepatic veins)

Grade VI

- Liver avulsion

Pancreas (Coccolini et al., World J Emerg Surg 2019, 14: 56):
Grade I
- Hematoma: Minor contusion without duct injury
- Laceration: Superficial laceration without duct injury

Grade II
- Hematoma: Major contusion without duct injury or tissue loss
- Laceration: Major laceration without duct injury or tissue loss

Grade III
- Laceration: Distal transection or parenchymal injury with duct injury

Grade IV
- Laceration: Proximal transection or parenchymal injury involving ampulla

Grade V
- Laceration: Massive disruption of pancreatic head

144:
The following is the American Association for Surgery of Trauma (AAST) renal injury scale (Coccolini et al., World J Emerg Surg 2019, 14: 54):

Grade I
- Contusion: Microscopic or gross hematuria
- Hematoma: Subcapsular, non-expanding without parenchymal laceration

Grade II
- Hematoma: Non-expanding perirenal hematoma confined to renal retroperitoneum
- Laceration: < 1 centimeter depth of renal cortex without urinary extravasation

Grade III
- Laceration: > 1 centimeter depth of renal cortex without collecting system rupture or urinary extravasation

Grade IV
- Laceration: Parenchymal laceration extending through renal cortex, medulla, and collecting system
- Vascular: Main renal artery or vein injury with contained hemorrhage

Grade V
- Laceration: Completely shattered kidney
- Vascular: Avulsion of renal hilum that devascularizes kidney

AAST grade IV-V blunt kidney injuries or any grade parenchymal injury with arterial dissection/occlusion is considered severe injury. Operative management is recommended for severe renal vascular injuries without self-limiting bleeding (Coccolini et al., World J Emerg Surg 2019, 14: 54). Since AAST grade IV-V are classified as severe renal injury, admission to the intensive care unit should be considered for close monitoring.

145:
Patients who suffer from pelvic fractures that disrupt the pelvic ring are susceptible to hemorrhage, which can result in hemodynamic instability. The disruption of the pelvic ring causes a decrease in the tamponade effect, leading to increased internal bleeding volume and further contributing to hemodynamic instability (Coccolini et al., World J Emerg Surg 2017, 12: 5). Two techniques that are commonly used to control bleeding in hemodynamically unstable patients with pelvic fractures are preperitoneal pelvic packing (PPP) and angioembolization (AE). Currently, there is no conclusive evidence to support either technique over the other (Bugaev et al., Am J Surg 2020, 220: 873-88). However, since venous bleeding sources are the main cause 80-90%(0.80-0.90) of hemorrhage, PPP

should be considered as the primary technique to control bleeding or for those who have persistent bleeding after AE. AE and PPP may be considered complementary procedures (Coccolini et al., World J Emerg Surg 2017, 12: 5).

146:

The modern method for preperitoneal pelvic packing involves a direct approach to the pelvis through a suprapubic midline incision, enabling access to the retroperitoneal space without violating the peritoneum (Skitch and Engels, Emerg Med Clin North Am 2018, 36: 161-79). This procedure can be performed in less than 20 minutes by an experienced clinician in either the emergency department or operating room (Coccolini et al., World J Emerg Surg 2017, 12: 5).

147:

In cases of unstable pelvic ring disruptions, external pelvic fixation can provide rigid temporary stability to the pelvic ring. External pelvic fixation is also recommended in conjunction with early hemorrhage management. If preperitoneal packing is used to control hemorrhage, it is recommended to place an external pelvic fixator to provide stable counterpressure for effective packing (Coccolini et al., World J Emerg Surg 2017, 12: 5).

148:

Bites from venomous snakes can result in envenomation. Toxins released may vary widely depending on the species of snakes in a specific geographic region. Toxins may be cytotoxic, causing pain and worsening edema; hematotoxic, causing consumption coagulopathy and bleeding; and/or neurotoxic, causing paralysis and respiratory insufficiency. Other symptoms that may be associated with snakebites include muscle necrosis, rhabdomyolysis, and acute kidney injury. The treatment of snake envenomation is the administration of antivenoms (Seifert et al., N Engl J Med 2022, 386: 68-78; Knudsen et al., Front Immunol 2021, 12: 661457).

149:

If clinical evidence of envenomation (e.g., neurotoxic, hepatotoxic, or cytotoxic symptoms) is present after a snakebite, antivenom should be administered as soon as possible. Early treatment with antivenom is the best approach to prevent death and disability resulting from a venomous snakebite. Although antivenom may not be readily available, every effort should be made to acquire it as quickly as possible while providing supportive care. The type of antivenom used depends on the species of the snake and the toxin released in the bite. Patients should be closely monitored for any worsening of symptoms and response to antivenom. This monitoring may include vital signs, urine output, neurological assessments, and serial lab tests. Depending on how the patient responds to the initial dose, additional doses of antivenom may be necessary (Benjamin at al., J Spec Oper Med 2020, 20: 43-74).

150:

Although open surgical repair has been traditionally used to treat peripheral vascular trauma, endovascular repair is now becoming increasingly popular. The placement of stent-grafts through endovascular intervention has been reported to be successful in managing injuries to isolated proximal extremity vascular injuries. However, endovascular intervention should not be used as the sole treatment method if there are signs of compartment syndrome, concerns about compression of a neurovascular bundle, or skin ischemia or necrosis, in which case, surgery would be necessary. After endovascular treatment, frequent neurovascular exams are recommended as distal embolization may occur following endovascular revascularization therapies (Weaver et al., Semin Intervent Radiol 2021, 38: 64-74).

151:

Social workers and case managers play an important role in assessing for suspected abuse of older individuals, initiating interventions, and assisting with safe discharge planning (American College of Surgeons Trauma Quality Programs, 2019 [cited Jun 27 2023]).

152:

All suspected cases of abuse or neglect of older individuals should be reported, as healthcare professionals are mandated to report such events in most US states. A reasonable suspicion of abuse is the only requirement for reporting, and most states do not require the consent of the victim. Adult Protective Services (APS) is the agency that investigates abuse or neglect in adults living in the community and in facilities (e.g., nursing homes, assisted living facilities) (American College of Surgeons Trauma Quality Programs, 2019 [cited Jun 27 2023]).

153:

Evidence supports nonoperative management (NOM) of hemodynamically stable patients with intra-abdominal organ injury if there are no other indications for laparotomy. NOM may consist of close observation, serial exams, repeated laboratory studies, vital sign monitoring, and supportive care (Coccolini et al., World J Emerg Surg 2020, 15: 24; Coccolini et al., World J Emerg Surg 2019, 14: 56; Coccolini et al., World J Emerg Surg 2019, 14: 54; Cimbanassi et al., J Trauma Acute Care Surg 2018, 84: 517-31; Coccolini et al., World J Emerg Surg 2017, 12: 40).

154:

Individuals with abdominal organ injuries treated with nonoperative management may initially be ordered for nothing by mouth (NPO) in case immediate surgical or endovascular intervention is required. In addition, some individuals may require bowel rest as a result of their injury (e.g., duodenal wall hematomas, traumatic pancreatitis) (Pavlidis et al., World J Gastrointest Surg 2022, 14: 538-43; Coccolini et al., World J Emerg Surg 2019, 14: 56).

155:

Hemodynamically stable patients with urinary tract injuries, including kidney, bladder, ureter and/or urethra, may be managed nonoperatively with urinary drainage using a foley or suprapubic catheter, placement of ureteral stents, or nephrostomy tube (Morey et al., J Urol 2021, 205: 30-5; Coccolini et al., World J Emerg Surg 2019, 14: 54; Yeung et al., J Trauma Acute Care Surg 2019, 86: 326-36; Morey et al., J Urol 2014, 192: 327-35).

156:

Respiratory interventions to prevent pulmonary complications in individuals with chest wall fractures may include (Dennis et al., Surg Clin North Am 2017, 97: 1047-64):

- Incentive spirometry
- Positive expiratory pressure (PEP) therapy
- Secretion clearance (e.g., mechanical insufflation-exsufflation (also known as cough assist), airway suctioning)

157:

Instruction: This criterion applies to patients with a CSF leak due to a head injury (e.g., closed skull fracture) that is being managed conservatively (e.g., bed rest). Those requiring a lumbar drain should be reviewed at the Critical level of care in this subset, while those requiring surgical repair should be reviewed in the General Surgical subset.

158:

Patients with mild hypothermia (89.6°F(32.0°C) to 95.0°F(35.0°C)) may or may not require active warming or passive warming. Moderate to severe hypothermia (less than 89.6°F(32.0°C)) usually requires active warming. Passive warming techniques include removing wet clothes and from a cold environment, applying insulation foils and blankets, active movement, and increasing ambient temperature. Active warming techniques include heat packs, forced air surface rewarming, warmed IV fluids, warm lavage of gastric, peritoneal, bladder, or thoracic space, and extracorporeal membrane oxygenation or cardiopulmonary bypass rewarming (Paal et al., Int J Environ Res Public Health 2022, 19: epub; van Veelen and Brodmann Maeder, Int J Environ Res Public Health 2021, 18: epub).

159:

In patients with blunt chest trauma, a recent study found that if only one of the measurements, either electrocardiography (ECG) or troponin is positive, the sensitivity of diagnosing a cardiac injury increases while the specificity decreases, compared to when both are positive (Kyriazidis et al., World J Emerg Surg 2023, 18: 36).

160:

A transthoracic echocardiogram (TTE) is recommended in the setting of suspected blunt cardiac injury when patients have an abnormal electrocardiogram (ECG) or elevated troponin to assess biventricular systolic function, evaluate for pericardial effusion, myocardial rupture, and/or septal and valvular injuries. Transesophageal echocardiography (TEE) may be considered if TTE findings are equivocal and aortic injury is suspected (Stojanovska et al., J Am Coll Radiol 2020, 17: S380-s90).

161:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.
For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.
NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.
(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

162:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

163:

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360: j5790).

164:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

165:

When foreign bodies cannot be retrieved endoscopically, observation may be necessary for certain foreign bodies (e.g., sharp pointed objects, batteries), and repeat radiographic imaging may be conducted to assess the object's passage through the gastrointestinal tract (Birk et al., Endoscopy 2016, 48: 489-96).

166:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

167:

Functional impairment refers to any condition that results in activity limitations impacting the patient's ability to ambulate, transfer, and/or perform activities of daily living (ADLs) or instrumental activities of daily living (IADLs). The patient will require assistance with one or more limitations.

168:

Instruction: These criteria refer to a functional impairment that develops during the course of this hospitalization.

169:

Acute kidney injury (AKI) is a term used to describe the broad spectrum of kidney function impairment, including insufficiency, oliguria, and failure. The spectrum ranges from minor changes in renal function markers (e.g., increase in serum creatinine, decreased urine output) to failure requiring renal replacement therapy.
AKI is defined as a sudden decrease in renal function resulting from multiple etiologies, including intrinsic kidney disease, ischemia, nephrotoxicity, and extrarenal pathology.
AKI encompasses patients with chronic kidney disease who experience an acute deterioration. AKI is associated with significant morbidity and mortality in hospitalized patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138; Wang et al., Am J Nephrol 2012, 35: 349-55).

170:

When the baseline serum creatinine level is unknown, the Modification of Diet in Renal Disease (MDRD) equation can be used to estimate an expected creatinine value (assuming a normal baseline value for glomerular filtration rate (GFR) of 75ml/min per 1.73 meters squared). The following values demonstrate one method to estimate baseline serum creatinine (eSCr) (with the omission of the race variable) (Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138; Bellomo et al., Crit Care 2004, 8: R204-12):
Male sex assigned at birth:

- Age 20-24 - eSCr 1.3 mg/dL (115 µmol/l)
- Age 25-39 - eSCr 1.2 mg/dL (106 µmol/l)
- Age 40-65 - eSCr 1.1 mg/dL (97 µmol/l)
- Age > 65 - eSCr 1.0 mg/dL (88 µmol/l)

Female sex assigned at birth:

- Age 20-29 - eSCr 1.0 mg/dL (88 µmol/l)
- Age 30-54 - eSCr 0.9 mg/dL (80 µmol/l)
- Age > 54 - eSCr 0.8 mg/dL (71 µmol/l)

Both the MDRD and the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPl) equations provide a calculation of estimated GFR using serum creatinine, sex assigned at birth, age, and race. Recognition of substantial diversity within population groups has led to multiple organizations no longer including race in the estimated GFR calculations. This approach was investigated by a joint task force of the National Kidney Foundation and the American Society of Nephrology, the result of which was a recommendation to implement the new CKD-EPI creatinine equation which does not use the race variable (Delgado et al., Am J Kidney Dis 2022, 79: 268-88 e1; Inker et al., N Engl J Med 2021, 385: 1737-49).

171:

This line of criteria refers to a patient without chronic kidney disease and an unknown creatinine baseline.

172:

Volume expanders are fluids administered intravenously to increase circulatory volume. Studies have demonstrated that a balanced crystalloid solution (e.g., Lactated Ringer's) is preferable to colloids in restoration of intravascular volume in sepsis and septic shock (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Winters et al., J Emerg Med 2017, 53: 928-39). However, crystalloids have been found to be less effective than colloids at stabilizing hemodynamic endpoints in critically ill patients (Martin and Bassett, J Crit Care 2019, 50: 144-54). The type of fluid selected for administration should be based on the indication for its use and other patient-specific factors.
Instruction: In order to apply criteria for volume expanders, there should be documentation of a volume deficit supported by clinical findings. The volume of infusion is patient-specific and varies based on the cause of volume depletion, comorbid condition, and patient response. Criteria for volume expander should not be applied for maintenance intravenous fluids or electrolyte replacement.

173:

For expansion of intravascular volume in patients with acute kidney injury (AKI), guidelines recommend initial treatment with an isotonic crystalloid solution (sodium chloride or sodium bicarbonate); the effects of sodium chloride on renal outcomes are being studied. Forced diuresis is not recommended because it can worsen AKI in volume-responsive patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Pistolesi et al., J Nephrol 2018, 31: 797-812; Joannidis et al., Intensive Care Med 2017, 43: 730-49; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

174:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

175:

Nephrotoxic agents include nonsteroidal anti-inflammatory drugs (NSAIDs), aminoglycosides, amphotericin B, high doses of loop diuretics, and anti-viral agents or other drugs (e.g., acyclovir and foscarnet) (Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

176:

Initial course refers to the initiation of a chronic hemodialysis/peritoneal dialysis program or initiation for acute renal pathology. A repeat initiation for a recurrence or exacerbation of an acute renal problem, provided that there was a gap of more than 14 days between treatments, would also qualify as an initial course.

177:

Following the discontinuation of dialysis, lab monitoring is necessary for continued management and may include blood urea nitrogen (BUN), creatinine, and electrolytes.

178:

The American College of Surgeons describes four classes of hemorrhage to assist with the early identification of hemorrhagic shock. Class II (mild) hemorrhage occurs when there is 15%-30%(0.15-0.30) blood volume loss and may manifest clinically as tachycardia. At this classification, a blood transfusion may be needed (Galvagno et al., Anesthesiol Clin 2019, 37: 13-32).

179:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements

repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).
Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

180:

Rehabilitation evaluation includes one or more of the following:

- Physical therapy (PT) evaluation and training
- Occupational therapy (OT) evaluation and training
- Speech therapy (ST) evaluation and training
- Cognitive evaluation and retraining
- Swallowing evaluation and retraining

181:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. These rates should be increased in the presence of dehydration or insensible loss. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

182:

Interdisciplinary care plays an important role in improving outcomes for older adults with frailty. The introduction of a frailty pathway on a trauma service that included consults to a geriatrician, physical therapist, nutritionist, and social worker, in addition to medication review and management, addressing healthcare goals, delirium prevention strategies, and fall prevention education, resulted in a significant decrease in delirium rates and readmissions within 30 days for individuals with frailty (Bryant et al., J Am Coll Surg 2019, 228: 852-9.e1).

183:

Specialty consultation refers to one or more of the following services:

- Geriatrician
- Nutrition
- Palliative care
- Pharmacy
- Social work or case management

184:

A serious complication of traumatic open-globe injuries is the development of post-traumatic endophthalmitis, which is an infection that affects the intraocular cavities. The presence of an intraocular foreign body, especially if made of organic material, delayed presentation more than 24 hours after the injury, and injuries in zone 1 (cornea and limbus) are associated with a higher risk of developing endophthalmitis (Keil et al., Clin Ophthalmol 2022, 16: 1401-11; Watanachai et al., J Ophthalmic Inflamm Infect 2021, 11: 22).

185:

In cases of open globe injuries, including those involving intraocular foreign bodies (IOFB), broad-spectrum intravenous (IV) anti-infectives are commonly administered upon presentation to minimize the risk of infection. The recommended anti-infective regimen typically includes IV vancomycin and third-generation cephalosporin or fluoroquinolones for patients with penicillin allergies. Following surgery, patients may require continued hospitalization for IV prophylactic anti-infectives (Keil et al., Clin Ophthalmol 2022, 16: 1401-11; Zhou et al., Clin Ophthalmol 2022, 16: 2545-59). Although there is a lack of evidence on the optimal duration of IV prophylactic anti-infectives, it has been recommended to continue IV anti-infectives for at least 48 hours post-operatively (Zhou et al., Clin Ophthalmol 2022, 16: 2545-59).

186:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

187:

Instruction: Patients with a scheduled designated inpatient surgery within 24 hours who require fluid management may be reviewed at the Acute level of care on the Pre-op day in the General Surgical subset.

188:

Chest tubes are used to evacuate air, fluid, or blood from the pleural space. For malignancies involving the pleura, chest tubes may also be used for the administration of sclerosing agents. Serial chest x-rays may be performed to ensure that there is no re-accumulation of air or fluid. The chest tube may be removed when there is no evidence of air leak, output has slowed to 250-400 mL/day, and the lung is fully expanded (Vuong et al., Respir Med 2018, 137: 152-66).

189:

Instruction: For this criteria point, repositioning refers to adjustment of the chest tube.

190:

Instruction: Patients with burns undergoing one or multiple debridement(s) or graft(s) in the operating room should continue to be reviewed in the General Trauma subset, as these procedures are part of burn wound therapy. The intent is for patients with burns to be managed in this subset which contains the interventions needed to appropriately care for this patient population.

191:

Although the vast majority of wound care is readily handled in the post-acute setting, there are situations when a patient requires care at the acute level, such as high-frequency wound care required, prolonged length of time required performing the procedure, and the type or amount of medication required to keep the patient comfortable during the procedure. For children less than 12 years of age (or less than 30 kg), the wound care regimen may need to be adjusted to a lesser frequency and/or duration (e.g., twice per day, 10 minutes), though still requiring the current level of care, due to the patient's size, pain tolerance, fluid shifts, and vital sign lability.
To determine the appropriate discharge destination, an assessment is required of the patient and/or caregiver's ability to learn the necessary wound care regimen, complexity of the treatment regimen, and a plan for managing contributory factors. For discharge to the home setting, the patient and/or caregiver must be able and willing to participate in the care plan, or there is sufficient agency assistance available to adequately meet the patient's needs. Patient and/or caregiver instruction should include all the following (Bryant and Nix, Acute & chronic wounds: current management concepts, 5th ed. 2015, xiv, 655 p.):

- Wound care regimen that includes a plan to monitor patient-specific contributory factors
- Adjunctive therapies (medications, nutritional plan, and supplements)
- Re-ordering medications and wound management supplies
- When to seek medical care (e.g., signs, symptoms, and complications to report to the health care provider)
- Continued follow-up and support with health care providers

192:

Wound care teaching should include instruction on the following:

- Adjunctive therapies (medications, nutritional plan, and supplements)
- Follow-up plan of care
- Medications and wound management supplies
- When to seek medical care (e.g., signs, symptoms, and complications to report to the health care provider)
- Wound care regimen (including patient-specific factors)

193:

Instruction: Examples of antiarrhythmics with proarrhythmic risk include, but are not limited to, amiodarone, disopyramide, dofetilide, dronedarone, flecainide, mexiletine, propafenone, and sotalol. Many factors influence the appropriate choice of drug, including underlying medical conditions and concurrent drug administration. In some instances, such as patients who are pregnant or are breastfeeding, there are drugs which are contraindicated or have limited evidence to support their use.

194:

Instruction: Patients with an electrical injury who are stepping down from the Critical level of care for continued cardiac monitoring can be reviewed here.

195:

Neurological impairment may include one or more of the following:

- Altered level of consciousness
- Cognitive deficit
- Motor deficit
- Speech deficit
- Sensory deficit

196:

Hyperkalemia may be associated with weakness leading to muscle twitching, paresthesia, and paralysis.

197:

Refractory arrhythmias, also known as uncontrolled, intractable, or drug-resistant arrhythmias, are resistant to treatment and are poorly controlled with medication. Recurrent arrhythmias are arrhythmias that occur more than once and may occur without warning.

198:

Blood pressure is considered labile when it is extremely variable and difficult to control.

199:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.
These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., Kidney Int 2020, 98: 294-309; Depret et al., Ann Intensive Care 2019, 9: 32; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

200:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

201:

Guidelines recommend monitoring patients for at least 24 hours after all symptoms have resolved post-snakebite treatment. Repeat labs are advised to ensure the resolution of coagulopathy (Benjamin at al., J Spec Oper Med 2020, 20: 43-74).

202:

The CMS Two-Midnight Rule generally requires that the medical record support a patient’s need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.
For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.
Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status;** those who do not meet criteria would be appropriate for secondary review.
(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

203:

Functional impairments may be safely managed in the home environment when the patient is willing and able to perform self-care independently with or without assistive devices, has sufficient caregiver support available to adequately meet the patient's needs or home care services are arranged.

204:

Initiation of enteral nutrition refers to the introduction of fluid into the gastrointestinal tract via oral intake or gastrointestinal tube feeding (e.g., G-tube, J-tube).

205:

In most patients hospitalized with infection, clinical improvement would be expected to occur within 2 days. For patients who are deteriorating or not responding to treatment, it may be necessary to broaden anti-infective coverage while awaiting the results of additional diagnostic tests. In addition, consultation by an infectious disease specialist and further diagnostic tests may be valuable to evaluate reasons for the lack of improvement.

206:

A drug-induced fever is a non-infectious febrile response following the administration of a drug that resolves when the drug is discontinued. The patient is often well-appearing and demonstrates resolution of the underlying disease process, despite the presence of a fever. A drug-induced fever should be suspected when the cause of the fever remains unknown, despite a full diagnostic work-up, or when the patient is receiving a drug frequently associated with fever. The most common agents triggering fever include anticonvulsants, anti-infectives, cardiac medications, allopurinol, and heparin.
The following drugs are also capable of inducing hyperthermia (Walter and Carraretto, Journal of the Intensive Care Society 2015):

- Serotonin-noradrenalin reuptake inhibitors
- Selective serotonin reuptake inhibitors
- Monoamine oxidase inhibitors
- Tricyclic antidepressants
- Bupropion
- Opioids
- Central nervous system stimulants
- Anticholinergics
- Antipsychotics
- Antispasmodics
- Sympathomimetic agents

Treatment involves discontinuing medications that are unnecessary, recently initiated, or highly associated with inducing fever. Resolution of the fever should occur within 48 to 72 hours of discontinuing the precipitating agent (Patel and Gallagher, Pharmacotherapy 2010, 30: 57-69; Eyer and Zilker, Crit Care 2007, 11: 236).

207:

For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

208:

Minor aortic injuries (e.g., small intimal tear, intramural hematoma) managed nonoperatively may require follow-up imaging. Some experts suggest repeating the computed tomography (CT) angiography 48-72 hours post injury to ensure the absence of injury progression requiring surgical intervention (Kapoor et al., Radiographics 2020, 40: 1834-47).

209:

Instruction: Patients with uncontrolled pain requiring continued stay for pain management can be reviewed under the General header using the “pain, uncontrolled” criteria.

210:

Instruction: This criterion is intended for patients who are post external pelvic fixator on Episode Day 1 for hemorrhage control and are now hemodynamically stable to step down to the Acute level of care.

211:

Respiratory interventions include airway clearance techniques (e.g., mechanical insufflation-exsufflation (also known as cough assist)), positive expiratory pressure (PEP) therapy, percussive intrapulmonary ventilation (IPV), chest physiotherapy (manual or by external oscillation device), suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

212:

Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

213:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

214:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

215:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

216:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

217:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

218:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

219:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

220:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.
Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

221:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

222:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

223:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

224:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

225:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.
The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

226:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

227:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

228:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

229:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

230:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.